

MATH 551: NUMERICAL ORDINARY DIFFERENTIAL EQUATIONS

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1. AUGUST 24, 2026

1.1. **Ordinary differential equations.** An ODE has the form

$$x'(t) = f(t, x(t)).$$

Given an initial condition $x(t_0) = x_0$, the corresponding *initial-value problem* (IVP) is to find $x(t)$ for $t \geq t_0$ such that

$$(1) \quad x'(t) = f(t, x(t)), \quad x(t_0) = x_0.$$

Topics arising in the numerical solution of differential equations include

- non-stiff problems and stiff solvers;
- geometric solvers;
- parameter estimation and Bayesian approaches, where the vector field may depend on a model parameter θ , as in $f(t, x, \theta)$;
- boundary-value problems; and
- stochastic differential equations (SDEs).

We generally consider vector-valued solutions and vector fields,

$$x: \mathbb{R} \longrightarrow \mathbb{R}^n, \quad f: \mathbb{R} \times \mathbb{R}^n \longrightarrow \mathbb{R}^n,$$

which can be written componentwise as

$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}, \quad f(t, x) = \begin{bmatrix} f_1(t, x_1, \dots, x_n) \\ \vdots \\ f_n(t, x_1, \dots, x_n) \end{bmatrix}.$$

1.2. Examples.

Example 1. *The scalar equation*

$$x' = x$$

is an ODE with $n = 1$.

Example 2 (Damped oscillator). *Consider the second-order equation*

$$x'' = -\omega^2 x - x'.$$

Setting $x_1 = x$ and $x_2 = x'$ converts it into the first-order system

$$\begin{cases} x'_1 = x_2, \\ x'_2 = -\omega^2 x_1 - x_2. \end{cases}$$

Thus, with $\mathbf{x} = (x_1, x_2)^\top$, we have $\mathbf{x}' = \mathbf{f}(x_1, x_2)$.

Example 3 (Chemical kinetics). *A three-species kinetic model is*

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \quad \begin{cases} x'_1 = -k_1x_1 + k_2x_2x_3, \\ x'_2 = k_1x_1 - k_2x_2x_3 - k_3x_2^2, \\ x'_3 = k_3x_2^2, \end{cases}$$

where $k_1, k_2, k_3 > 0$ are reaction rates. The reaction rates may occur on very different scales, which can lead to stiffness.

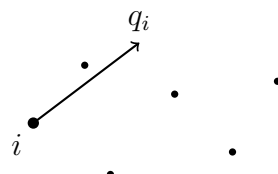
Example 4 (Molecular model). *Let q_i be the position of particle i , let m_i be its mass, and let $V(q_1, \dots, q_N)$ be the potential energy. Newton's equation is*

$$m_i q''_i = -\frac{\partial V(q_1, \dots, q_N)}{\partial q_i}.$$

Introducing the momentum $p_i = m_i q'_i$ gives

$$\begin{cases} q'_i = \frac{p_i}{m_i}, \\ p'_i = -\frac{\partial V}{\partial q_i}. \end{cases}$$

The positions and momenta can therefore be collected into the state vector

$$\mathbf{x} = \begin{bmatrix} q_1 \\ p_1 \\ q_2 \\ p_2 \\ \vdots \\ q_N \\ p_N \end{bmatrix}.$$


1.3. Solutions and the Lipschitz condition. Consider the IVP (1) on an interval containing t_0 .

Definition 1. A function $x(t)$ is a solution of the IVP if it satisfies the integral equation

$$x(t) = x_0 + \int_{t_0}^t f(s, x(s)) ds.$$

Definition 2 (Lipschitz condition). The vector field f is Lipschitz continuous in x on a domain D if there is a constant $L \geq 0$ such that

$$\|f(t, x) - f(t, y)\| \leq L\|x - y\|$$

for all $x, y \in D$ and all relevant t .

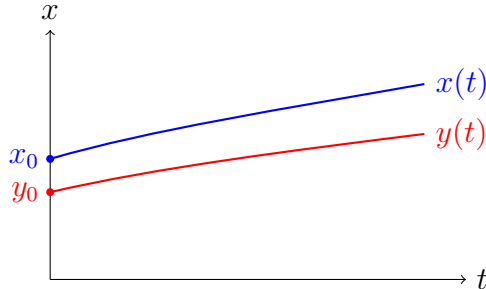
Under the usual continuity assumption in t , the Lipschitz condition in x implies that the solution of the IVP exists and is unique on an appropriate interval.

The same condition also controls the sensitivity to the initial data. Let $x(t)$ and $y(t)$ be solutions with initial values x_0 and y_0 , respectively. Their integral equations give

$$\begin{aligned} \|x(t) - y(t)\| &\leq \|x_0 - y_0\| + \int_{t_0}^t \|f(s, x(s)) - f(s, y(s))\| ds \\ &\leq \|x_0 - y_0\| + L \int_{t_0}^t \|x(s) - y(s)\| ds. \end{aligned}$$

By Grönwall's inequality,

$$(2) \quad \|x(t) - y(t)\| \leq \|x_0 - y_0\| e^{L(t-t_0)}.$$



Thus nearby initial values produce solutions whose separation is bounded by (2).

2. AUGUST 26, 2026

The main topics are properties of solutions, time discretization, and consistency error. We continue to consider the IVP

$$(3) \quad x'(t) = f(t, x(t)), \quad x(t_0) = x_0,$$

where $x: \mathbb{R} \rightarrow \mathbb{R}^n$.

2.1. Stability of solutions. The integral formulation of (3) is

$$x(t) = x_0 + \int_{t_0}^t f(s, x(s)) ds.$$

If $y(t)$ is another solution with initial value y_0 , then

$$y(t) = y_0 + \int_{t_0}^t f(s, y(s)) ds,$$

and hence

$$y(t) - x(t) = y_0 - x_0 + \int_{t_0}^t (f(s, y(s)) - f(s, x(s))) ds.$$

Suppose f is Lipschitz continuous in x with Lipschitz constant L , so that

$$\|f(t, y) - f(t, x)\| \leq L\|y - x\|.$$

Taking norms gives

$$(4) \quad \|y(t) - x(t)\| \leq \|y_0 - x_0\| + \int_{t_0}^t L\|y(s) - x(s)\| ds.$$

The following inequality controls this accumulation of error.

Lemma 1 (Grönwall's inequality). *Let $u, v \in C([t_0, t_1], \mathbb{R})$ be nonnegative functions and let $\rho \geq 0$. If*

$$(5) \quad u(t) \leq \rho + \int_{t_0}^t v(s)u(s) ds, \quad t \in [t_0, t_1],$$

then

$$(6) \quad u(t) \leq \rho \exp \left(\int_{t_0}^t v(s) ds \right), \quad t \in [t_0, t_1].$$

In particular, $u \equiv 0$ when $\rho = 0$.

Proof. First suppose that $\rho > 0$ and define

$$U(t) = \rho + \int_{t_0}^t v(s)u(s) ds.$$

Then $U \in C^1([t_0, t_1], \mathbb{R})$, and

$$u(t) \leq U(t), \quad U(t) \geq \rho > 0.$$

Since u and v are nonnegative,

$$U'(t) = v(t)u(t) \leq v(t)U(t).$$

Thus the logarithmic derivative of U satisfies

$$(\log U(t))' = \frac{U'(t)}{U(t)} \leq v(t).$$

Integrating from t_0 to t yields

$$\log U(t) - \log U(t_0) \leq \int_{t_0}^t v(s) ds.$$

Because $U(t_0) = \rho$ and $u(t) \leq U(t)$, applying the exponential function proves (6) when $\rho > 0$.

Now suppose that $\rho = 0$. For any $\varepsilon > 0$, (5) implies

$$u(t) \leq \varepsilon + \int_{t_0}^t v(s)u(s) ds.$$

The first part of the proof, with $\rho = \varepsilon$, gives

$$0 \leq u(t) \leq \varepsilon \exp \left(\int_{t_0}^t v(s) ds \right).$$

Letting $\varepsilon \downarrow 0$ shows that $u(t) = 0$ for every $t \in [t_0, t_1]$. □

Apply Lemma 1 to (4) with

$$u(t) = \|y(t) - x(t)\|, \quad v(t) = L, \quad \rho = \|y_0 - x_0\|.$$

We obtain

$$(7) \quad \|y(t) - x(t)\| \leq \|y_0 - x_0\| e^{L(t-t_0)}.$$

This estimate can be pessimistic, but it establishes continuous dependence on the initial data.

2.2. The solution map. The *solution map*, or *flow map*, is denoted by

$$x(t) = \Phi^{t,t_0}(x_0).$$

For example, consider the scalar logistic equation

$$x' = x(1 - x).$$

For $x_0 \neq 0$, its solution is

$$\Phi^{t,t_0}(x_0) = \frac{1}{1 + \frac{1-x_0}{x_0}e^{-(t-t_0)}}.$$

When $x_0 = 0$, the solution is $x(t) \equiv 0$.

The solution map has the identity and composition properties

$$\begin{aligned}\Phi^{t_0,t_0}(x_0) &= x_0, & \Phi^{t_0,t_0} &= I, \\ \Phi^{t_2,t_1}(\Phi^{t_1,t_0}(x_0)) &= \Phi^{t_2,t_0}(x_0).\end{aligned}$$

The second property is the basis of time marching.

Differentiating the solution map with respect to time gives

$$\begin{aligned}\frac{d}{dt}\Phi^{t,t_0}(x_0) &= f(t, x(t)), \\ \frac{d^2}{dt^2}\Phi^{t,t_0}(x_0) &= f_t(t, x(t)) + f_x(t, x(t))x'(t) \\ &= (f_t + f_x f)(t, x(t)),\end{aligned}$$

where f_x is the Jacobian matrix of f with respect to x . Higher time derivatives can be found by repeated differentiation, for example with automatic differentiation; they can also be viewed through the Koopman operator.

To measure sensitivity with respect to the initial value, define the Wronskian matrix

$$W(t) = \frac{\partial}{\partial x_0}\Phi^{t,t_0}(x_0) = \frac{\partial x(t)}{\partial x_0}.$$

Differentiating with respect to t gives

$$W'(t) = \frac{\partial}{\partial x_0}f(t, x(t))$$

$$= f_x(t, x(t)) \frac{\partial x(t)}{\partial x_0}.$$

Thus, if

$$A(t) = f_x(t, x(t)),$$

then W satisfies the *variational equation*

$$(8) \quad W'(t) = A(t)W(t), \quad W(t_0) = I.$$

2.3. One-step methods. Choose a time grid

$$t_0 < t_1 < \cdots < t_N = t, \quad \tau_j = t_{j+1} - t_j,$$

where the step sizes τ_j are small. Repeated use of the composition property gives

$$\Phi^{t_N, t_0} = \Phi^{t_N, t_{N-1}} \circ \cdots \circ \Phi^{t_2, t_1} \circ \Phi^{t_1, t_0}.$$

It is therefore enough to approximate a single short-time solution map.

Writing $t_j = t$ and $t_{j+1} = t + \tau$, Taylor expansion gives

$$(9) \quad \Phi^{t+\tau, t}(x) = x + \tau f(t, x) + \frac{\tau^2}{2} (f_t + f_x f)(t, x) + O(\tau^3).$$

Dropping the terms of order τ^2 and higher defines the Euler one-step map

$$\Psi^{t+\tau, t}(x) = x + \tau f(t, x).$$

Consequently, Euler's method is

$$(10) \quad x_{j+1} = x_j + \tau_j f(t_j, x_j).$$

Its one-step consistency error follows from (9):

$$\Phi^{t+\tau, t}(x) - \Psi^{t+\tau, t}(x) = \frac{\tau^2}{2} (f_t + f_x f)(t, x) + O(\tau^3) = O(\tau^2).$$

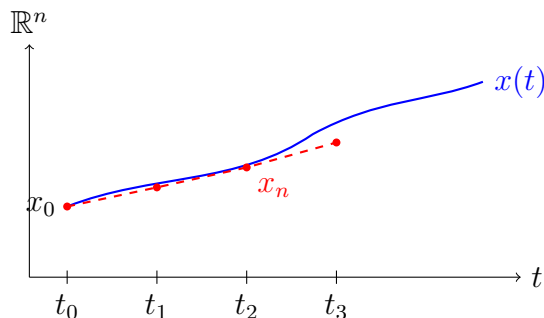
3. AUGUST 28, 2026

We continue the study of one-step methods, focusing on consistency, examples, and convergence. Let

$$t_0 < t_1 < \cdots < t_N, \quad \tau_n = t_{n+1} - t_n,$$

where t_n is a time level and τ_n is the corresponding step size. We use $x(t_n)$ for the exact solution and x_n for its numerical approximation:

$$x(t_n) \approx x_n.$$



A one-step method advances the numerical solution by a map

$$(11) \quad x_{n+1} = \Psi^{t_{n+1}, t_n}(x_n).$$

Thus the exact and numerical solutions at t_n can be written as

$$\begin{aligned} x(t_n) &= \Phi^{t_n, t_{n-1}} \circ \dots \circ \Phi^{t_2, t_1} \circ \Phi^{t_1, t_0}(x_0), \\ x_n &= \Psi^{t_n, t_{n-1}} \circ \dots \circ \Psi^{t_2, t_1} \circ \Psi^{t_1, t_0}(x_0). \end{aligned}$$

3.1. Methods obtained from quadrature. The integral form of the ODE over one step is

$$(12) \quad x(t_{n+1}) = x(t_n) + \int_{t_n}^{t_{n+1}} f(s, x(s)) ds.$$

Different quadrature approximations of this integral produce different one-step methods.

3.1.1. Forward Euler method. Using the left endpoint in (12) gives

$$x(t_{n+1}) \approx x(t_n) + \tau_n f(t_n, x(t_n)).$$

Replacing the exact solution values by numerical approximations yields

$$(13) \quad x_{n+1} = x_n + \tau_n f(t_n, x_n) = \Psi_{\text{FE}}^{t_{n+1}, t_n}(x_n).$$

3.1.2. Backward Euler method. Using the right endpoint instead gives

$$x(t_{n+1}) \approx x(t_n) + \tau_n f(t_{n+1}, x(t_{n+1})).$$

The resulting method is

$$(14) \quad x_{n+1} = x_n + \tau_n f(t_{n+1}, x_{n+1}) = \Psi_{\text{BE}}^{t_{n+1}, t_n}(x_n).$$

Because x_{n+1} occurs on both sides, backward Euler requires the solution of a fixed-point problem at each step.

3.1.3. *A predictor–corrector method.* The trapezoidal rule suggests

$$x(t_{n+1}) \approx x(t_n) + \frac{\tau_n}{2} (f(t_n, x(t_n)) + f(t_{n+1}, x(t_{n+1}))).$$

We can first predict the right endpoint with forward Euler and then use that prediction in the trapezoidal formula:

$$(15) \quad x_{n+1}^* = x_n + \tau_n f(t_n, x_n),$$

$$(16) \quad x_{n+1} = x_n + \frac{\tau_n}{2} (f(t_n, x_n) + f(t_{n+1}, x_{n+1}^*)).$$

This explicit two-stage scheme is also called Heun’s method. Together, the two stages define a one-step map

$$x_{n+1} = \Psi_{\text{H}}^{t_{n+1}, t_n}(x_n).$$

3.2. **Local consistency error.** For a numerical one-step map Ψ , define its *local error*, or *consistency error*, at (t_n, x) by

$$(17) \quad \mathcal{E}_n(x) = \Phi^{t_{n+1}, t_n}(x) - \Psi^{t_{n+1}, t_n}(x).$$

Here the exact and numerical steps start from the same value x at time t_n .

Example 5 (Forward Euler). *The Taylor expansion from (9) gives*

$$\begin{aligned} \mathcal{E}_n^{\text{FE}}(x) &= \Phi^{t_n + \tau_n, t_n}(x) - (x + \tau_n f(t_n, x)) \\ &= \frac{\tau_n^2}{2} (f_t + f_x f)(t_n, x) + O(\tau_n^3) \\ &= O(\tau_n^2). \end{aligned}$$

Example 6 (Backward Euler). *Let*

$$y(\tau) = \Psi_{\text{BE}}^{t_n + \tau, t_n}(x).$$

The defining fixed-point equation is

$$y(\tau) = x + \tau f(t_n + \tau, y(\tau)).$$

Differentiating this equation and evaluating at $\tau = 0$ gives

$$y'(0) = f(t_n, x), \quad y''(0) = 2(f_t + f_x f)(t_n, x).$$

Therefore,

$$\Psi_{\text{BE}}^{t_n+\tau_n, t_n}(x) = x + \tau_n f(t_n, x) + \tau_n^2 (f_t + f_x f)(t_n, x) + O(\tau_n^3).$$

Comparing this with the exact flow expansion yields

$$\begin{aligned} \mathcal{E}_n^{\text{BE}}(x) &= -\frac{\tau_n^2}{2} (f_t + f_x f)(t_n, x) + O(\tau_n^3) \\ &= O(\tau_n^2). \end{aligned}$$

3.3. Global error and convergence. The *global error* at t_n is

$$(18) \quad e_n = x_n - x(t_n).$$

The local errors accumulate as the method marches forward in time. On a fixed time interval there are approximately $1/\tau$ steps of size τ , so one generally loses one power of τ when passing from local to global error:

$$\mathcal{E}_n = O(\tau^{p+1}) \implies e_n = O(\tau^p).$$

More generally, with

$$h = \max_{0 \leq n < N} \tau_n,$$

a stable one-step method with local error $O(h^{p+1})$ has global error $O(h^p)$ under the usual smoothness assumptions. Thus both forward and backward Euler have local error of order two and global order one.

4. AUGUST 31, 2026

We continue the analysis of one-step methods. The main goals are to pass from local consistency to global convergence and to introduce Runge–Kutta methods. Consider the IVP

$$x'(t) = f(t, x(t)), \quad x(t_0) = x_0 \in \mathbb{R}^n,$$

on a grid

$$t_0 < t_1 < \dots < t_N = T, \quad \tau_j = t_{j+1} - t_j, \quad h = \max_{0 \leq j < N} \tau_j.$$

4.1. From local consistency to global convergence. Recall that Φ^{t_{j+1}, t_j} denotes the exact flow over one step, while Ψ^{t_{j+1}, t_j} denotes the numerical one-step map. For a starting value x , define the local consistency error by

$$(19) \quad \xi_j(x) = \Phi^{t_{j+1}, t_j}(x) - \Psi^{t_{j+1}, t_j}(x).$$

Definition 3 (Consistency order). *A one-step method has consistency order $p > 0$ if, uniformly for the relevant starting values,*

$$\|\xi_j(x)\| \leq C\tau_j^{p+1}$$

as $h \rightarrow 0$. The order p need not be an integer when the data have only Hölder regularity.

Forward and backward Euler have consistency order one. The exact and numerical values at the final time can be written as compositions:

$$\begin{aligned} x(T) &= \Phi^{t_N, t_{N-1}} \circ \dots \circ \Phi^{t_2, t_1} \circ \Phi^{t_1, t_0}(x_0), \\ x_N &= \Psi^{t_N, t_{N-1}} \circ \dots \circ \Psi^{t_2, t_1} \circ \Psi^{t_1, t_0}(x_0). \end{aligned}$$

Thus the local errors are propagated through all the subsequent steps.

Let

$$e_j = x_j - x(t_j)$$

be the global error. For brevity, set $\Psi_j = \Psi^{t_{j+1}, t_j}$ and $\Phi_j = \Phi^{t_{j+1}, t_j}$.

Then

$$\begin{aligned} e_{j+1} &= \Psi_j(x_j) - \Phi_j(x(t_j)) \\ (20) \quad &= \Psi_j(x_j) - \Psi_j(x(t_j)) - \xi_j(x(t_j)). \end{aligned}$$

Suppose the one-step map satisfies the stability estimate

$$(21) \quad \|\Psi_j(y) - \Psi_j(z)\| \leq e^{L\tau_j} \|y - z\|.$$

Combining the consistency and stability estimates in (20) gives

$$(22) \quad \|e_{j+1}\| \leq e^{L\tau_j} \|e_j\| + C\tau_j^{p+1}.$$

Since $e_0 = 0$, iteration yields

$$\begin{aligned}\|e_N\| &\leq C \sum_{j=0}^{N-1} \tau_j^{p+1} e^{L(T-t_{j+1})} \\ &\leq Ch^p \sum_{j=0}^{N-1} \tau_j e^{L(T-t_{j+1})}.\end{aligned}$$

For $L > 0$, the sum is bounded by a geometric integral, and hence

$$(23) \quad \|e_N\| \leq \frac{C}{L} (e^{L(T-t_0)} - 1) h^p.$$

When $L = 0$, the corresponding bound is

$$\|e_N\| \leq C(T - t_0) h^p.$$

In either case, $h \rightarrow 0$ implies $e_N \rightarrow 0$. Thus stability converts a local error of order $p + 1$ into a global error of order p .

Remark 1 (Contractive problems). In the favorable case

$$\|\Psi_j(y) - \Psi_j(z)\| \leq \|y - z\|,$$

no exponential amplification occurs. Direct summation gives

$$\|e_N\| \leq C \sum_{j=0}^{N-1} \tau_j^{p+1} \leq C(T - t_0) h^p.$$

Such contractivity is especially relevant for dissipative ODEs and numerical methods that preserve their dissipative structure.

4.2. Runge–Kutta methods. Consider an explicit method with s stages and coefficients a_{ij} , b_i , and c_i . Starting from x at time t with step size τ , its stages are computed successively from

$$(24) \quad k_i = f \left(t + c_i \tau, x + \tau \sum_{j=1}^{i-1} a_{ij} k_j \right), \quad i = 1, \dots, s,$$

and its one-step map is

$$(25) \quad \Psi^{t+\tau, t}(x) = x + \tau \sum_{i=1}^s b_i k_i.$$

Example 7 (Forward Euler). *Take $s = 1$, $c_1 = 0$, $a_{11} = 0$, and $b_1 = 1$. Then*

$$k_1 = f(t, x), \quad \Psi^{t+\tau, t}(x) = x + \tau k_1,$$

which is forward Euler, a first-order method.

Example 8 (Explicit trapezoidal method). *Take $s = 2$ and*

$$A = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \quad b = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix}, \quad c = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

The stages and update are

$$\begin{aligned} k_1 &= f(t, x), \\ k_2 &= f(t + \tau, x + \tau k_1), \\ \Psi^{t+\tau, t}(x) &= x + \frac{\tau}{2}(k_1 + k_2). \end{aligned}$$

This is the predictor–corrector method from the previous lecture, also called Heun’s method. It has order two.

5. SEPTEMBER 2, 2026

We develop the coefficient representation of Runge–Kutta methods and derive their order conditions. Throughout, we retain the notation of the previous lectures: $\tau_n = t_{n+1} - t_n$, x_n is the numerical approximation at t_n , Φ is the exact flow, and Ψ is the numerical one-step map.

5.1. Butcher tableaux. An s -stage Runge–Kutta method has the form

$$(26) \quad k_i = f \left(t_n + c_i \tau_n, x_n + \tau_n \sum_{j=1}^s a_{ij} k_j \right), \quad i = 1, \dots, s,$$

$$(27) \quad x_{n+1} = x_n + \tau_n \sum_{i=1}^s b_i k_i.$$

The coefficients are collected into

$$A = (a_{ij})_{i,j=1}^s, \quad b = \begin{bmatrix} b_1 & \cdots & b_s \end{bmatrix}^\top, \quad c = \begin{bmatrix} c_1 & \cdots & c_s \end{bmatrix}^\top.$$

They are conventionally displayed in a *Butcher tableau*:

$$\begin{array}{c|cccc} c_1 & a_{11} & a_{12} & \cdots & a_{1s} \\ c_2 & a_{21} & a_{22} & \cdots & a_{2s} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ c_s & a_{s1} & a_{s2} & \cdots & a_{ss} \\ \hline & b_1 & b_2 & \cdots & b_s \end{array}$$

The values $t_n + c_i\tau_n$ are the intermediate stage times between t_n and t_{n+1} . If A is strictly lower triangular, then each stage depends only on previously computed stages. The resulting method is an *explicit Runge–Kutta method*.

5.2. The internal consistency relation. Any nonautonomous ODE can be converted into an autonomous system by augmenting its state. Set

$$z(t) = \begin{bmatrix} t \\ x(t) \end{bmatrix}, \quad z'(t) = \begin{bmatrix} 1 \\ f(t, x(t)) \end{bmatrix} = F(z(t)).$$

Apply a Runge–Kutta method to this autonomous system. Because the time component of every stage slope is one, the time component of stage i is

$$t_n + \tau_n \sum_{j=1}^s a_{ij}.$$

For this to agree with the prescribed stage time $t_n + c_i\tau_n$ in (26), we require

$$c_i = \sum_{j=1}^s a_{ij}, \quad i = 1, \dots, s.$$

Thus the *internal consistency relation* is

$$(28) \quad c = A\mathbf{1}, \quad \mathbf{1} = \begin{bmatrix} 1 & \cdots & 1 \end{bmatrix}^T.$$

Under this relation, applying the Runge–Kutta scheme directly to the nonautonomous equation is equivalent to applying it to the augmented autonomous system.

5.3. One-step error and order. For an exact starting value x at time t , define the one-step defect by

$$(29) \quad \mathcal{E}(x, t, \tau) = \Phi^{t+\tau, t}(x) - \Psi^{t+\tau, t}(x).$$

A Runge–Kutta method has order p if

$$\mathcal{E}(x, t, \tau) = O(\tau^{p+1}) \quad \text{as } \tau \rightarrow 0.$$

Assuming (28), it suffices to derive the conditions for an autonomous equation

$$x' = f(x).$$

We write its exact and Runge–Kutta flows as Φ^τ and Ψ^τ , respectively.

Let $f' = Df(x)$ and $f'' = D^2f(x)$. Differentiation along an exact solution gives

$$\begin{aligned} x' &= f, \\ x'' &= f'f, \\ x''' &= f''(f, f) + f'f'f. \end{aligned}$$

Consequently,

$$(30) \quad \Phi^\tau(x) = x + \tau f + \frac{\tau^2}{2} f'f + \frac{\tau^3}{6} (f''(f, f) + f'f'f) + O(\tau^4).$$

For the numerical method, set

$$\Delta_i = \tau \sum_{j=1}^s a_{ij} k_j, \quad k_i = f(x + \Delta_i).$$

Since $k_j = f + O(\tau)$ and $c = A\mathbf{1}$,

$$\Delta_i = \tau c_i f + O(\tau^2).$$

A more detailed Taylor expansion gives

$$k_i = f + \tau c_i f'f + \tau^2 \left(\sum_{j=1}^s a_{ij} c_j f'f'f + \frac{c_i^2}{2} f''(f, f) \right) + O(\tau^3).$$

Substitution into $\Psi^\tau(x) = x + \tau \sum_i b_i k_i$ yields

$$\Psi^\tau(x) = x + \tau(b^\top \mathbf{1})f + \tau^2(b^\top c)f'f$$

$$(31) \quad + \tau^3 \left((b^\top A c) f' f' f + \frac{1}{2} (b^\top c^{\circ 2}) f''(f, f) \right) + O(\tau^4),$$

where $c^{\circ 2} = (c_1^2, \dots, c_s^2)^\top$.

Comparing (30) and (31) gives the order conditions through order three:

$$(32) \quad b^\top \mathbf{1} = 1, \quad (\text{order one}),$$

$$(33) \quad b^\top c = \frac{1}{2}, \quad (\text{additional condition for order two}),$$

$$(34) \quad b^\top c^{\circ 2} = \frac{1}{3}, \quad (\text{additional condition for order three}),$$

$$(35) \quad b^\top A c = \frac{1}{6}, \quad (\text{additional condition for order three}).$$

Therefore (32) gives a first-order method; the first two conditions give a second-order method; and all four give a third-order method. A fourth-order method must satisfy eight conditions in total. For nonautonomous equations, the relation $c = A\mathbf{1}$ is also needed for these autonomous order conditions to apply directly.

5.4. Examples.

Example 9 (Forward Euler). *Forward Euler has the tableau*

$$\begin{array}{c|c} 0 & 0 \\ \hline & 1 \end{array}.$$

Thus $s = 1$, $A = 0$, $b = 1$, and $c = 0$. It satisfies $b^\top \mathbf{1} = 1$, but not the second-order condition, so it is first order.

Example 10 (Explicit trapezoidal method). *The explicit trapezoidal method, or Heun's method, is*

$$\begin{aligned} k_1 &= f(t_n, x_n), \\ k_2 &= f(t_n + \tau_n, x_n + \tau_n k_1), \\ x_{n+1} &= x_n + \frac{\tau_n}{2} (k_1 + k_2). \end{aligned}$$

Its tableau is

$$\begin{array}{c|cc} 0 & 0 & 0 \\ 1 & 1 & 0 \\ \hline & \frac{1}{2} & \frac{1}{2} \end{array}.$$

Here $c = A\mathbf{1}$, $b^\top \mathbf{1} = 1$, and $b^\top c = 1/2$. However, $b^\top c^{\circ 2} = 1/2 \neq 1/3$, so the method is second order but not third order.

5.5. Explicit three-stage methods. A general explicit three-stage Runge–Kutta method has tableau

$$\begin{array}{c|ccc} 0 & 0 & 0 & 0 \\ c_2 & a_{21} & 0 & 0 \\ c_3 & a_{31} & a_{32} & 0 \\ \hline & b_1 & b_2 & b_3 \end{array}.$$

Internal consistency requires

$$c_2 = a_{21}, \quad c_3 = a_{31} + a_{32}.$$

Before imposing the order conditions, the six free coefficients are

$$b_1, b_2, b_3, a_{21}, a_{31}, a_{32}.$$

The four conditions (32)–(35) enforce third order, so one generically expects a two-parameter family of explicit three-stage schemes of order three.

6. SEPTEMBER 4, 2026

We continue the study of Runge–Kutta methods, first constructing families of explicit methods and then introducing embedded methods and stability functions. We use the notation established in the previous lectures: $\tau_n = t_{n+1} - t_n$, x_n is the numerical solution, and Φ and Ψ denote the exact and numerical flows.

6.1. Runge–Kutta methods and order conditions. Recall that an s -stage Runge–Kutta method is

$$k_i = f \left(t_n + c_i \tau_n, x_n + \tau_n \sum_{j=1}^s a_{ij} k_j \right), \quad i = 1, \dots, s,$$

$$x_{n+1} = x_n + \tau_n \sum_{i=1}^s b_i k_i.$$

Here s is the number of stages, $A = (a_{ij})$ contains the stage coefficients, c_i is the i -th stage node, and b_i is the i -th quadrature weight. The internal consistency relation is

$$c = A\mathbf{1}.$$

It makes the method agree on the original nonautonomous equation and its autonomous augmentation. The method is explicit precisely when A is strictly lower triangular.

For the autonomous equation $x' = f(x)$, a method has order p when

$$\|\Phi^\tau(x) - \Psi^\tau(x)\| = O(\tau^{p+1}) \quad \text{as } \tau \rightarrow 0.$$

The first four order conditions, derived in the previous lecture, are

$$\begin{aligned} b^\top \mathbf{1} &= 1, & b^\top c &= \frac{1}{2}, \\ b^\top c^{\circ 2} &= \frac{1}{3}, & b^\top Ac &= \frac{1}{6}. \end{aligned}$$

They are sufficient through order three. To obtain order four, four more conditions are needed:

$$\begin{aligned} b^\top c^{\circ 3} &= \frac{1}{4}, & b^\top (c \circ (Ac)) &= \frac{1}{8}, \\ b^\top A(c^{\circ 2}) &= \frac{1}{12}, & b^\top A^2 c &= \frac{1}{24}, \end{aligned}$$

where $\circ$ denotes componentwise multiplication. Thus one, two, four, and eight conditions must be satisfied for orders one, two, three, and four, respectively.

Forward Euler uses one stage and has order one, while the explicit trapezoidal method uses two stages and has order two. We next continue the construction of higher-order explicit methods.

6.2. Explicit three-stage methods of order three. For an explicit three-stage method, internal consistency gives the tableau

$$\begin{array}{c|ccc} 0 & 0 & 0 & 0 \\ c_2 & a_{21} & 0 & 0 \\ c_3 & a_{31} & a_{32} & 0 \\ \hline & b_1 & b_2 & b_3 \end{array}, \quad c_2 = a_{21}, \quad c_3 = a_{31} + a_{32}.$$

There are six free coefficients and four order conditions, so the third-order methods generically form a two-parameter family. Let

$$c_2 = \alpha, \quad c_3 = \beta.$$

Solving the four order conditions gives

$$\begin{array}{c|ccc} 0 & 0 & 0 & 0 \\ \alpha & \alpha & 0 & 0 \\ \beta & \frac{\beta(3\alpha(1-\alpha)-\beta)}{\alpha(2-3\alpha)} & \frac{\beta(\beta-\alpha)}{\alpha(2-3\alpha)} & 0 \\ \hline & \frac{6\alpha\beta-3\alpha-3\beta+2}{6\alpha\beta} & \frac{3\beta-2}{6\alpha(\beta-\alpha)} & \frac{2-3\alpha}{6\beta(\beta-\alpha)} \end{array},$$

provided the displayed denominators are nonzero.

A useful subfamily has $\beta = 1$. Its weights are

$$b_1 = \frac{3\alpha-1}{6\alpha}, \quad b_2 = \frac{1}{6\alpha(1-\alpha)}, \quad b_3 = \frac{2-3\alpha}{6(1-\alpha)}.$$

All three weights are nonnegative when $\alpha \in [1/3, 2/3]$, but positivity of the weights alone does not guarantee that one method is better than another.

Example 11 (Kutta's third-order method). *Taking $\alpha = 1/2$ and $\beta = 1$ gives*

$$\begin{array}{c|ccc} 0 & 0 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 1 & -1 & 2 & 0 \\ \hline & \frac{1}{6} & \frac{2}{3} & \frac{1}{6} \end{array}.$$

Thus

$$k_1 = f(t_n, x_n),$$

$$\begin{aligned}k_2 &= f\left(t_n + \frac{\tau_n}{2}, x_n + \frac{\tau_n}{2}k_1\right), \\k_3 &= f\left(t_n + \tau_n, x_n + \tau_n(-k_1 + 2k_2)\right), \\x_{n+1} &= x_n + \frac{\tau_n}{6}(k_1 + 4k_2 + k_3).\end{aligned}$$

6.3. Explicit four-stage methods of order four. Before imposing internal consistency, a general explicit four-stage tableau contains thirteen coefficients: three nontrivial nodes, six strictly lower-triangular entries of A , and four weights. Internal consistency removes three degrees of freedom, and the eight fourth-order conditions generically leave two free parameters.

In the usual nonconfluent family one may take

$$c_1 = 0, \quad c_2 = \alpha, \quad c_3 = \beta, \quad c_4 = 1,$$

with distinct stage nodes. The generic formulas exclude degenerate parameter choices; one convenient nondegeneracy condition is

$$\alpha\beta(\alpha - 1)(\beta - 1)(\alpha - \beta)(2\alpha - 1)(6\alpha\beta - 4\alpha - 4\beta + 3) \neq 0.$$

Example 12 (The three-eighths rule). *Choosing $\alpha = 1/3$ and $\beta = 2/3$ produces the fourth-order three-eighths rule*

$$\begin{array}{c|cccc}0 & 0 & 0 & 0 & 0 \\ \frac{1}{3} & \frac{1}{3} & 0 & 0 & 0 \\ \frac{2}{3} & -\frac{1}{3} & 1 & 0 & 0 \\ 1 & 1 & -1 & 1 & 0 \\ \hline & \frac{1}{8} & \frac{3}{8} & \frac{3}{8} & \frac{1}{8}\end{array}.$$

Its stages are

$$\begin{aligned}k_1 &= f(t_n, x_n), \\k_2 &= f\left(t_n + \frac{\tau_n}{3}, x_n + \frac{\tau_n}{3}k_1\right), \\k_3 &= f\left(t_n + \frac{2\tau_n}{3}, x_n + \tau_n\left(-\frac{1}{3}k_1 + k_2\right)\right), \\k_4 &= f\left(t_n + \tau_n, x_n + \tau_n(k_1 - k_2 + k_3)\right), \\x_{n+1} &= x_n + \frac{\tau_n}{8}(k_1 + 3k_2 + 3k_3 + k_4).\end{aligned}$$

Repeated nodes give a separate, *confluent* family. Setting $c_2 = c_3 = 1/2$ and letting $\delta = a_{32} \neq 0$ yields

$$(36) \quad \begin{array}{c|cccc} 0 & 0 & 0 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} - \delta & \delta & 0 & 0 \\ 1 & 0 & 1 - \frac{1}{2\delta} & \frac{1}{2\delta} & 0 \\ \hline & \frac{1}{6} & \frac{4\delta - 1}{6\delta} & \frac{1}{6\delta} & \frac{1}{6} \end{array}.$$

Example 13 (Classical RK4). *The choice $\delta = 1/2$ in (36) gives the classical tableau*

$$\begin{array}{c|cccc} 0 & 0 & 0 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ \hline & \frac{1}{6} & \frac{1}{3} & \frac{1}{3} & \frac{1}{6} \end{array}.$$

Equivalently,

$$\begin{aligned} k_1 &= f(t_n, x_n), \\ k_2 &= f\left(t_n + \frac{\tau_n}{2}, x_n + \frac{\tau_n}{2}k_1\right), \\ k_3 &= f\left(t_n + \frac{\tau_n}{2}, x_n + \frac{\tau_n}{2}k_2\right), \\ k_4 &= f(t_n + \tau_n, x_n + \tau_n k_3), \\ x_{n+1} &= x_n + \frac{\tau_n}{6}(k_1 + 2k_2 + 2k_3 + k_4). \end{aligned}$$

6.4. Embedded pairs. Adaptive solvers need local error estimates. They obtain them by combining the same stages in two different ways. Given fixed A and c , choose two weight vectors b and $\widehat{b}$ and form

$$\begin{aligned} x_{n+1}^{[p]} &= x_n + \tau_n \sum_{i=1}^s b_i k_i, \\ x_{n+1}^{[p+1]} &= x_n + \tau_n \sum_{i=1}^s \widehat{b}_i k_i. \end{aligned}$$

The two formulas share all evaluations k_i , so

$$\text{err}_n = \|x_{n+1}^{[p+1]} - x_{n+1}^{[p]}\|$$

is an inexpensive local error estimate. If this quantity is well below the requested tolerance, the next step size can be increased. If it is too large, the step size is decreased and the step may be rejected.

MATLAB's `ode45` uses a seven-stage Dormand–Prince embedded pair of orders four and five. The minimal number of stages for an explicit Runge–Kutta method grows with its order; the low-order values discussed in class are

p	1	2	3	4	5	6	7	8
$s_{\min}$	1	2	3	4	6	7	9	11

6.5. The stability function. Apply a Runge–Kutta method to the linear test equation

$$(37) \quad y'(t) = \lambda y(t).$$

With $z = \tau_n \lambda$, one numerical step has the form

$$(38) \quad y_{n+1} = R(z)y_n,$$

where R is the *stability function*. The exact amplification factor is e^z .

For a Runge–Kutta method, the stage equations applied to (37) give

$$(I - zA)k = \lambda y_n \mathbf{1}.$$

Substituting the resulting stages into the update gives

$$(39) \quad R(z) = 1 + zb^T(I - zA)^{-1}\mathbf{1}.$$

If the method has order p , then necessarily

$$(40) \quad R(z) - e^z = O(z^{p+1}) \quad \text{as } z \rightarrow 0.$$

For example, the eight fourth-order conditions imply

$$R(z) = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \frac{z^4}{4!} + O(z^5).$$

Condition (40) is necessary for order p , but it only tests a linear ODE and is not sufficient to establish order p for general nonlinear equations.

7. SEPTEMBER 9, 2026

This lecture introduces extrapolation as a way to increase the order of an existing time-stepping method. We then begin the study of stability as a property of solutions of the differential equation itself.

7.1. Accuracy and computational cost. Consider a method of order p on the uniform grid

$$t_n = t_0 + n\tau, \quad n = 0, \dots, N, \quad t_N = T,$$

and let $Y_\tau(T) = x_N$ denote the numerical approximation at the final time. Convergence of order p means that, for sufficiently small τ ,

$$(41) \quad \|Y_\tau(T) - x(T)\| \leq C(T)\tau^p.$$

In particular, $Y_\tau(T) \rightarrow x(T)$ as $\tau \rightarrow 0$. For a fixed step size, increasing the order generally decreases the error, provided that τ is in the asymptotic regime in which (41) is valid.

Suppose a tolerance $\varepsilon > 0$ is prescribed. Balancing the error estimate with the tolerance gives

$$C(T)\tau^p \approx \varepsilon, \quad \tau \approx \left(\frac{\varepsilon}{C(T)} \right)^{1/p}.$$

Since $N = (T - t_0)/\tau$, the resulting work, measured by the number of time steps, is approximately

$$(42) \quad N \approx (T - t_0) \left(\frac{C(T)}{\varepsilon} \right)^{1/p}.$$

Thus a higher-order method has a more favorable dependence on the tolerance. Formula (42) does not include the work per step or the size of the error constant, both of which also matter in practice.

7.2. Richardson extrapolation. Extrapolation requires more than an error bound. Assume the stronger asymptotic expansion

$$(43) \quad Y_\tau(T) = x(T) + c_p(T)\tau^p + c_{p+1}(T)\tau^{p+1} + c_{p+2}(T)\tau^{p+2} + \dots$$

Rerunning the same method with half the step size gives

$$Y_{\tau/2}(T) = x(T) + c_p(T)\frac{\tau^p}{2^p} + c_{p+1}(T)\frac{\tau^{p+1}}{2^{p+1}} + O(\tau^{p+2}).$$

Multiplying this equation by 2^p and subtracting (43) cancels the leading error term:

$$2^p Y_{\tau/2}(T) - Y_\tau(T) = (2^p - 1)x(T) + O(\tau^{p+1}).$$

Consequently, the *Richardson extrapolate*

$$(44) \quad Y_\tau^{[1]}(T) = \frac{2^p Y_{\tau/2}(T) - Y_\tau(T)}{2^p - 1}$$

satisfies

$$Y_\tau^{[1]}(T) = x(T) + O(\tau^{p+1}).$$

This gains one order. The underlying one-step method is unchanged; the additional accuracy comes from combining two complete runs.

7.2.1. Symmetric methods. For an autonomous problem, a one-step method is *symmetric* if a step of length $-\tau$ exactly reverses a step of length τ :

$$\Psi^{-\tau} = (\Psi^\tau)^{-1}.$$

Symmetric methods have even order. Under the usual smoothness assumptions, their fixed-time global error expansion contains powers of one parity. If the method has order p , the expansion takes the form

$$(45) \quad Y_\tau(T) = x(T) + d_p(T)\tau^p + d_{p+2}(T)\tau^{p+2} + d_{p+4}(T)\tau^{p+4} + \dots$$

Applying the same combination as in (44) cancels the τ^p term, but now there is no τ^{p+1} term. Hence

$$\frac{2^p Y_{\tau/2}(T) - Y_\tau(T)}{2^p - 1} = x(T) + O(\tau^{p+2}),$$

so one extrapolation raises the order by two.

7.3. Extrapolation from several step sizes. The preceding cancellation can be repeated using several numerical solutions. Choose distinct positive integers $n_1, \dots, n_m$, set

$$\tau_i = \frac{\bar{\tau}}{n_i}, \quad i = 1, \dots, m,$$

and compute $Y_{\tau_i}(T)$ for every i . Motivated by (43), fit the generalized polynomial

$$(46) \quad P(h) = a_0 + \sum_{j=0}^{m-2} a_{p+j} h^{p+j}$$

to these values:

$$(47) \quad P(\tau_i) = Y_{\tau_i}(T), \quad i = 1, \dots, m.$$

The unknowns $a_0, a_p, \dots, a_{p+m-2}$ are determined by the linear system

$$\begin{bmatrix} 1 & \tau_1^p & \tau_1^{p+1} & \dots & \tau_1^{p+m-2} \\ 1 & \tau_2^p & \tau_2^{p+1} & \dots & \tau_2^{p+m-2} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & \tau_m^p & \tau_m^{p+1} & \dots & \tau_m^{p+m-2} \end{bmatrix} \begin{bmatrix} a_0 \\ a_p \\ a_{p+1} \\ \vdots \\ a_{p+m-2} \end{bmatrix} = \begin{bmatrix} Y_{\tau_1}(T) \\ Y_{\tau_2}(T) \\ \vdots \\ Y_{\tau_m}(T) \end{bmatrix}.$$

For a vector-valued ODE, the same coefficient matrix is used for every component. Extrapolating to zero step size gives

$$(48) \quad \bar{x}(T) = P(0) = a_0.$$

The first $m - 1$ error terms have been removed, so, under the assumed full expansion,

$$\bar{x}(T) = x(T) + O(\bar{\tau}^{p+m-1}).$$

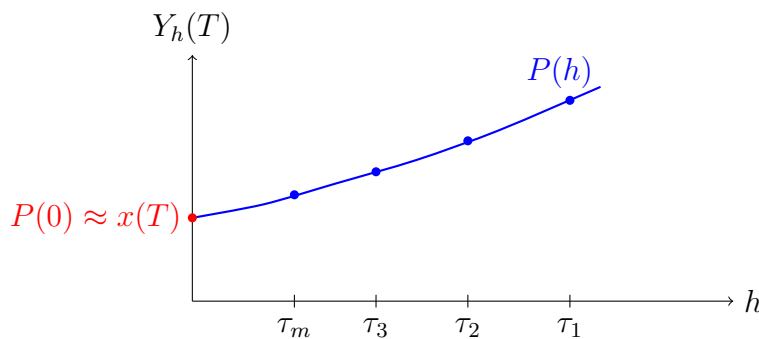
Equivalently, a_0 is a linear combination of the computed values,

$$a_0 = \sum_{i=1}^m \gamma_i Y_{\tau_i}(T),$$

where the weights are chosen so that

$$\sum_{i=1}^m \gamma_i = 1, \quad \sum_{i=1}^m \gamma_i \tau_i^{p+j} = 0, \quad j = 0, \dots, m-2.$$

For $m = 2$, $n_1 = 1$, and $n_2 = 2$, these conditions recover (44). For a symmetric error expansion, one instead uses the powers $p, p+2, p+4, \dots$ in (46).



7.4. Stability of solutions. Return to the IVP

$$(49) \quad x'(t) = f(t, x(t)), \quad x(t_0) = x_0,$$

and write its solution as $x(t; t_0, x_0)$ to display its dependence on the initial data.

Definition 4 (Stability of a solution). *The solution through x_0 is stable if, for every $\varepsilon > 0$, there exists $\delta > 0$ such that*

$$\|x_0 - y_0\| < \delta \implies \|x(t; t_0, x_0) - x(t; t_0, y_0)\| < \varepsilon$$

for every $t \geq t_0$.

The estimate from Grönwall's inequality in (7) gives continuous dependence over every fixed finite interval. Stability is stronger: the same δ must control the separation for all future times.

Definition 5 (Asymptotic stability). *A stable solution through x_0 is asymptotically stable if all sufficiently close initial values y_0 also satisfy*

$$\|x(t; t_0, x_0) - x(t; t_0, y_0)\| \longrightarrow 0 \quad \text{as } t \rightarrow \infty.$$

For an autonomous equation $x' = f(x)$, a point x_* satisfying

$$f(x_*) = 0$$

is an *equilibrium*. The solution starting at the equilibrium is constant: if $x_0 = x_*$, then $x(t; t_0, x_*) = x_*$ for all $t \geq t_0$. Stability of this equilibrium is obtained from the preceding definitions by comparing this constant solution with solutions starting at nearby points.

Example 14 (Linear autonomous system). *Consider*

$$x' = Ax.$$

The origin is an equilibrium, and

$$x(t; t_0, x_0) = e^{(t-t_0)A}x_0.$$

Thus the difference of two solutions is

$$x(t; t_0, x_0) - x(t; t_0, y_0) = e^{(t-t_0)A}(x_0 - y_0).$$

The system is stable precisely when e^{tA} remains bounded for $t \geq 0$, and it is asymptotically stable precisely when $e^{tA} \rightarrow 0$. In spectral terms, stability requires every eigenvalue to have nonpositive real part and every eigenvalue on the imaginary axis to be semisimple. Asymptotic stability holds exactly when every eigenvalue has strictly negative real part.

In the scalar case $x' = \lambda x$, the solution is $x(t) = e^{\lambda(t-t_0)}x_0$. Hence the equilibrium at the origin is stable when $\operatorname{Re} \lambda \leq 0$, and it is asymptotically stable when $\operatorname{Re} \lambda < 0$.

These notions describe the behavior of the exact differential equation. They are distinct from numerical absolute stability, which asks whether a time-stepping method reproduces the decay of a stable test equation.

8. SEPTEMBER 11, 2026

We continue the discussion of stability from the previous lecture. The main goals are to linearize an autonomous ODE near an equilibrium, to study the resulting system using the matrix exponential, and to extend the solution formula to time-dependent linear systems by means of the Dyson series.

8.1. Linearization near an equilibrium. Consider the autonomous nonlinear equation

$$(50) \quad x' = f(x)$$

and let x_* be an equilibrium, so that $f(x_*) = 0$. Introduce the perturbation

$$y = x - x_*.$$

Since x_* is constant, $y' = x'$. If f is continuously differentiable, its expansion near x_* is

$$f(x_* + y) = f(x_*) + f_x(x_*)y + r(y) = Ay + r(y), \quad A = f_x(x_*),$$

where

$$\frac{\|r(y)\|}{\|y\|} \longrightarrow 0 \quad \text{as } y \rightarrow 0.$$

Thus the exact perturbation equation is

$$(51) \quad y' = Ay + r(y),$$

and its linearization is

$$(52) \quad y' = Ay.$$

The equilibrium x_* corresponds to the zero solution of the perturbation equation. Hence studying whether $x(t)$ remains close to x_* begins with studying whether solutions of (52) remain close to zero.

8.2. Autonomous linear systems and the matrix exponential.

Consider the initial-value problem

$$(53) \quad y' = Ay, \quad y(t_0) = y_0,$$

where $A \in \mathbb{R}^{n \times n}$ is constant. Repeated differentiation gives

$$y^{(k)}(t_0) = A^k y_0.$$

The Taylor series about t_0 therefore gives

$$y(t) = y_0 + (t - t_0)Ay_0 + \frac{(t - t_0)^2}{2!}A^2y_0 + \cdots$$

$$= \left(\sum_{k=0}^{\infty} \frac{(t-t_0)^k A^k}{k!} \right) y_0.$$

Definition 6 (Matrix exponential). *For a square matrix B , its exponential is*

$$(54) \quad e^B = \sum_{k=0}^{\infty} \frac{B^k}{k!}.$$

The series converges in every matrix norm. With this notation, the solution of (53) is

$$(55) \quad y(t) = e^{(t-t_0)A} y_0.$$

Indeed,

$$\frac{d}{dt} e^{(t-t_0)A} = A e^{(t-t_0)A} = e^{(t-t_0)A} A, \quad e^{0A} = I.$$

As observed in the previous lecture, the zero solution of (53) is stable exactly when

$$(56) \quad \sup_{t \geq 0} \|e^{tA}\| < \infty,$$

and it is asymptotically stable exactly when

$$(57) \quad \|e^{tA}\| \longrightarrow 0 \quad \text{as } t \rightarrow \infty.$$

These statements follow directly from

$$y(t; t_0, y_0) - y(t; t_0, z_0) = e^{(t-t_0)A} (y_0 - z_0).$$

Linearity also implies that stability of the zero solution is equivalent to stability of every solution.

8.3. Time-dependent linear systems and the Dyson series. Now consider the nonautonomous linear equation

$$(58) \quad y' = A(t)y, \quad y(t_0) = y_0,$$

where $A(t)$ is continuous. Its integral equation is

$$(59) \quad y(t) = y_0 + \int_{t_0}^t A(t_1)y(t_1) dt_1.$$

At an intermediate time t_1 , the same equation reads

$$y(t_1) = y_0 + \int_{t_0}^{t_1} A(t_2)y(t_2) dt_2.$$

Substituting this formula into (59) gives

$$\begin{aligned} y(t) &= y_0 + \left(\int_{t_0}^t A(t_1) dt_1 \right) y_0 \\ &\quad + \int_{t_0}^t \int_{t_0}^{t_1} A(t_1)A(t_2)y(t_2) dt_2 dt_1. \end{aligned}$$

Continuing this substitution produces the *Dyson series*

$$(60) \quad y(t) = U(t, t_0)y_0,$$

where the evolution operator is

$$(61) \quad U(t, t_0) = I + \sum_{k=1}^{\infty} \int_{t_0}^t \int_{t_0}^{t_1} \cdots \int_{t_0}^{t_{k-1}} A(t_1)A(t_2) \cdots A(t_k) dt_k \cdots dt_2 dt_1.$$

If $\|A(s)\| \leq M$ on $[t_0, t]$, the norm of the k -th term is at most

$$\frac{M^k(t - t_0)^k}{k!},$$

which proves convergence of the series. Termwise differentiation shows that

$$\frac{\partial}{\partial t} U(t, t_0) = A(t)U(t, t_0), \quad U(t_0, t_0) = I.$$

Because matrices evaluated at different times need not commute, the order $t_1 \geq t_2 \geq \cdots \geq t_k$ in (61) is essential. A common shorthand is the *time-ordered exponential*

$$(62) \quad U(t, t_0) = \mathcal{T} \exp \left(\int_{t_0}^t A(s) ds \right).$$

If $A(s)A(r) = A(r)A(s)$ for every r, s , time ordering is unnecessary and

$$U(t, t_0) = \exp \left(\int_{t_0}^t A(s) ds \right).$$

In particular, when $A(t) \equiv A$, the ordered integration region has volume $(t - t_0)^k/k!$, and the Dyson series reduces to $e^{(t-t_0)A}$.

8.4. Properties of the matrix exponential. The power-series definition immediately gives several useful identities.

Proposition 1. *Let A and B be square matrices of the same size.*

(i) *If M is invertible, then*

$$e^{MAM^{-1}} = Me^AM^{-1}.$$

(ii) *If $AB = BA$, then*

$$e^{A+B} = e^Ae^B = e^Be^A.$$

(iii) *For a block-diagonal matrix,*

$$e^{\text{diag}(A_1, \dots, A_m)} = \text{diag}(e^{A_1}, \dots, e^{A_m}).$$

Proof. For the first identity, observe that $(MAM^{-1})^k = MA^kM^{-1}$ and apply (54). The third identity follows similarly because powers preserve block-diagonal structure. When A and B commute, the binomial theorem is valid for $(A+B)^k$. Substitution into the exponential series and collection of like terms gives the second identity. $\square$

8.5. Jordan form and linear stability. The preceding identities reduce the stability analysis of $y' = Ay$ to the analysis of individual Jordan blocks. Over $\mathbb{C}$, write

$$(63) \quad A = PJP^{-1}, \quad J = \text{diag}(J_1, \dots, J_m).$$

If J_k is a Jordan block of size q_k associated with the eigenvalue λ_k , then

$$J_k = \lambda_k I + N_k, \quad N_k^{q_k} = 0.$$

Since $\lambda_k I$ commutes with N_k ,

$$(64) \quad e^{tJ_k} = e^{t\lambda_k} \left(I + tN_k + \frac{t^2 N_k^2}{2!} + \dots + \frac{t^{q_k-1} N_k^{q_k-1}}{(q_k-1)!} \right).$$

Thus each Jordan block contributes an exponential factor multiplied by a polynomial in t .

Theorem 1 (Stability criterion for a linear system). *For $y' = Ay$, the following statements hold.*

(i) *The zero solution is asymptotically stable if and only if*

$$\operatorname{Re} \lambda < 0 \quad \text{for every } \lambda \in \sigma(A).$$

(ii) *The zero solution is stable if and only if every eigenvalue satisfies $\operatorname{Re} \lambda \leq 0$ and every eigenvalue on the imaginary axis is semisimple, meaning that all of its Jordan blocks have size one.*

Proof. By Proposition 1 and (63),

$$e^{tA} = P e^{tJ} P^{-1} = P \operatorname{diag}(e^{tJ_1}, \dots, e^{tJ_m}) P^{-1}.$$

If $\operatorname{Re} \lambda_k < 0$, the exponential decay in (64) dominates every polynomial factor, so the block tends to zero. If $\operatorname{Re} \lambda_k > 0$, the block has an exponentially growing solution. When $\operatorname{Re} \lambda_k = 0$, the scalar exponential has constant magnitude. The block is therefore bounded precisely when its nilpotent part is zero, or equivalently when its size is one. Combining the blocks with (56) and (57) proves both assertions. $\square$

Example 15 (A defective eigenvalue on the imaginary axis). *Let*

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

The only eigenvalue is zero, but its Jordan block has size two. Since $A^2 = 0$,

$$e^{tA} = I + tA = \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix}.$$

Hence

$$y_2(t) = y_2(0), \quad y_1(t) = y_1(0) + t y_2(0).$$

Arbitrarily small initial data with $y_2(0) \neq 0$ lead to linear growth, so the origin is not stable. This illustrates why nonpositive real parts alone are insufficient for stability.

For the nonlinear perturbation equation (51), the linearization principle gives two decisive cases. If the spectrum of $f_x(x_*)$ lies in the open left half-plane, then x_* is locally asymptotically stable. If the spectrum contains an eigenvalue with positive real part, then x_* is unstable. If it meets the imaginary axis but not the open right half-plane, linearization alone may be inconclusive; the nonlinear remainder $r(y)$ can determine the behavior.

9. SEPTEMBER 14, 2026

We develop two complementary approaches to stability. Linearization transfers spectral information from a Jacobian matrix to a nonlinear equilibrium, while Lyapunov functions establish stability without computing the solution explicitly. We then apply the same ideas to time-dependent linear equations and to the discrete evolution produced by Runge–Kutta methods.

9.1. From linear to nonlinear stability. Recall the autonomous equation

$$x' = f(x),$$

an equilibrium x_* satisfying $f(x_*) = 0$, and its Jacobian

$$A = f_x(x_*).$$

With $y = x - x_*$, the perturbation equation derived in (51) is

$$y' = Ay + r(y), \quad \frac{\|r(y)\|}{\|y\|} \longrightarrow 0 \quad \text{as } y \rightarrow 0.$$

The matrix A is called *Hurwitz* if all of its eigenvalues lie in the open left half-plane:

$$\operatorname{Re} \lambda < 0 \quad \text{for every } \lambda \in \sigma(A).$$

The linearization theorem gives the following nonlinear conclusion.

Theorem 2 (Linearization criterion). *Suppose f is continuously differentiable near x_* .*

- (i) *If $f_x(x_*)$ is Hurwitz, then x_* is locally asymptotically stable.*
- (ii) *If $f_x(x_*)$ has an eigenvalue with positive real part, then x_* is unstable.*

If no eigenvalue of $f_x(x_*)$ lies on the imaginary axis, the equilibrium is *hyperbolic*. The Hartman–Grobman theorem says that the nonlinear flow near a hyperbolic equilibrium is locally topologically equivalent to its linearization $y' = Ay$. If the spectrum meets the imaginary axis, the linearized system alone may not determine stability.

9.2. Lyapunov's direct method. Eigenvalue calculations or explicit solution formulas can be expensive. Lyapunov's method instead measures the size of a solution by a scalar energy-like function.

Definition 7 (Lyapunov function). *A continuously differentiable function V , defined near an equilibrium x_* , is a Lyapunov function if*

$$V(x_*) = 0, \quad V(x) > 0 \quad \text{for } x \neq x_*$$

near x_ , and its derivative along solutions,*

$$(65) \quad \dot{V}(x) = \nabla V(x)^\top f(x),$$

is nonpositive.

Theorem 3 (Lyapunov stability theorem). *If V is positive definite near x_* and $\dot{V} \leq 0$, then x_* is stable. If, in addition, $\dot{V}(x) < 0$ for every $x \neq x_*$ sufficiently close to x_* , then x_* is locally asymptotically stable.*

The idea is that a solution cannot cross outward through a sufficiently small level set of V when $V(x(t))$ is nonincreasing. Strict decrease forces the solution toward the equilibrium.

9.3. Quadratic Lyapunov functions for linear systems. Consider again

$$(66) \quad y' = Ay.$$

Let Q be a real symmetric positive-definite matrix and define

$$(67) \quad V(y) = y^\top Q y.$$

If $\lambda_{\min}(Q)$ and $\lambda_{\max}(Q)$ denote the extreme eigenvalues of Q , then

$$(68) \quad \lambda_{\min}(Q)\|y\|^2 \leq V(y) \leq \lambda_{\max}(Q)\|y\|^2.$$

Consequently, $V(y(t)) \rightarrow 0$ if and only if $y(t) \rightarrow 0$, and boundedness of $V(y(t))$ is equivalent to boundedness of $y(t)$.

Along a solution of (66),

$$\begin{aligned} \frac{d}{dt}V(y(t)) &= y'(t)^\top Q y(t) + y(t)^\top Q y'(t) \\ (69) \qquad \qquad &= y(t)^\top (A^\top Q + Q A) y(t). \end{aligned}$$

If

$$A^\top Q + Q A \preceq -\mu I \quad \text{for some } \mu > 0,$$

then

$$\begin{aligned} \dot{V}(y(t)) &\leq -\mu \|y(t)\|^2 \\ &\leq -\frac{\mu}{\lambda_{\max}(Q)} V(y(t)). \end{aligned}$$

Grönwall's inequality therefore gives

$$(70) \qquad V(y(t)) \leq e^{-\mu(t-t_0)/\lambda_{\max}(Q)} V(y(t_0)),$$

so the origin is exponentially stable.

9.3.1. *Lyapunov equation.* Choose a symmetric positive-definite matrix C . The matrix equation

$$(71) \qquad A^\top Q + Q A = -C$$

is called the *continuous Lyapunov equation*. Often one simply takes $C = I$ and solves for Q .

Theorem 4 (Lyapunov matrix theorem). *The matrix A is Hurwitz if and only if, for every symmetric $C \succ 0$, equation (71) has a unique symmetric solution $Q \succ 0$.*

Proof. Suppose first that A is Hurwitz. By the stability criterion from the previous lecture, $e^{tA} \rightarrow 0$ exponentially. Define

$$(72) \qquad Q = \int_0^\infty e^{tA^\top} C e^{tA} dt.$$

The integral converges, Q is symmetric, and for $v \neq 0$,

$$v^\top Q v = \int_0^\infty (e^{tA} v)^\top C (e^{tA} v) dt > 0.$$

Thus $Q \succ 0$. Moreover,

$$\begin{aligned} A^\top Q + QA &= \int_0^\infty \frac{d}{dt} \left(e^{tA^\top} C e^{tA} \right) dt \\ &= \left[e^{tA^\top} C e^{tA} \right]_{t=0}^{t=\infty} = -C. \end{aligned}$$

If Q_1 and Q_2 are two solutions, their difference D satisfies $A^\top D + DA = 0$. Hence $e^{tA^\top} D e^{tA} = D$ for all t , while the left-hand side tends to zero. Therefore $D = 0$, proving uniqueness.

Conversely, suppose $Q \succ 0$ and $C \succ 0$ satisfy (71). For a possibly complex eigenpair $Av = \lambda v$, multiplication by the conjugate transpose v^* gives

$$2 \operatorname{Re}(\lambda) v^* Q v = -v^* C v < 0.$$

Since $v^* Q v > 0$, every eigenvalue has negative real part. Thus A is Hurwitz. $\square$

The integral formula shows the equivalence between the spectral and Lyapunov approaches to linear stability. It also proves the nonlinear stability conclusion locally. Indeed, choose $C = I$ and use the same quadratic function for $y' = Ay + r(y)$. Then

$$\dot{V}(y) = -\|y\|^2 + 2y^\top Q r(y) = -\|y\|^2 + o(\|y\|^2),$$

which is negative for all sufficiently small nonzero y .

9.4. A time-dependent linear system. Consider

$$(73) \quad y' = A(t)y.$$

For the simple Lyapunov function

$$V(y) = \frac{1}{2} y^\top y = \frac{1}{2} \|y\|^2,$$

we have

$$(74) \quad \dot{V}(y(t)) = y(t)^\top S(t) y(t), \quad S(t) = \frac{1}{2} (A(t) + A(t)^\top).$$

Thus $S(t) \preceq 0$ for all $t \geq t_0$ is sufficient for stability, because the Euclidean norm cannot increase. The uniform condition

$$(75) \quad S(t) \preceq -\alpha I, \quad t \geq t_0, \quad \alpha > 0,$$

implies

$$\dot{V} \leq -2\alpha V, \quad \|y(t)\| \leq e^{-\alpha(t-t_0)}\|y(t_0)\|,$$

and hence uniform exponential stability. Pointwise negative definiteness without a uniform bound does not by itself guarantee decay to zero.

9.5. Runge–Kutta methods on a linear system. We now examine the numerical evolution of the constant-coefficient system $y' = Ay$ with a fixed step size τ .

9.5.1. *Forward Euler and an explicit midpoint method.* Forward Euler has one stage,

$$k_1 = Ay_n,$$

and therefore

$$(76) \quad y_{n+1} = y_n + \tau k_1 = (I + \tau A)y_n.$$

The two-stage explicit midpoint method is

$$\begin{aligned} k_1 &= Ay_n, \\ k_2 &= A \left(y_n + \frac{\tau}{2} k_1 \right), \\ y_{n+1} &= y_n + \tau k_2. \end{aligned}$$

Consequently,

$$(77) \quad y_{n+1} = \left(I + \tau A + \frac{\tau^2}{2} A^2 \right) y_n.$$

Every explicit two-stage Runge–Kutta method of order two has the same update matrix on this linear equation, even though its internal stages may differ.

More generally, recall the Runge–Kutta stability function R from (39). Applied to $y' = Ay$, a Runge–Kutta method gives

$$(78) \quad y_{n+1} = G(\tau A)y_n, \quad G(\tau A) = R(\tau A).$$

For an explicit s -stage method, R is a polynomial of degree at most s . If that method also has order s , then

$$R(z) = \sum_{j=0}^s \frac{z^j}{j!},$$

so $G(\tau A)$ is the degree- s Taylor approximation of $e^{\tau A}$.

9.6. Stability of a discrete linear evolution. Consider a general recurrence

$$(79) \quad y_{n+1} = G y_n, \quad y_n = G^n y_0.$$

The zero solution is stable precisely when $\{G^n\}_{n \geq 0}$ is bounded, and it is asymptotically stable precisely when $G^n \rightarrow 0$.

As in the continuous case, write $G = PJP^{-1}$ over $\mathbb{C}$. For a Jordan block

$$J_k = \mu_k I + N_k, \quad N_k^{q_k} = 0,$$

the binomial theorem gives

$$(80) \quad J_k^n = \sum_{j=0}^{\min(n, q_k-1)} \binom{n}{j} \mu_k^{n-j} N_k^j.$$

The binomial coefficient grows polynomially in n . It is dominated by $|\mu_k|^n$ when $|\mu_k| < 1$, but produces growth on the unit circle when $N_k \neq 0$.

Theorem 5 (Stability criterion for a discrete linear system). *For the recurrence (79):*

(i) *the zero solution is asymptotically stable if and only if*

$$\rho(G) = \max_{\mu \in \sigma(G)} |\mu| < 1;$$

(ii) *the zero solution is stable if and only if $\rho(G) \leq 1$ and every eigenvalue with $|\mu| = 1$ is semisimple.*

For a Runge–Kutta method, the spectral mapping theorem applied to (78) gives

$$\sigma(R(\tau A)) = \{R(\tau \lambda) : \lambda \in \sigma(A)\}.$$

Therefore the numerical solution is asymptotically stable when

$$(81) \quad |R(\tau\lambda)| < 1 \quad \text{for every } \lambda \in \sigma(A).$$

This motivates the *absolute stability region*

$$(82) \quad \mathcal{S} = \{z \in \mathbb{C} : |R(z)| < 1\}.$$

For example, forward Euler has $R(z) = 1 + z$. If $\lambda < 0$ is real, then $|1 + \tau\lambda| < 1$ is equivalent to

$$0 < \tau < \frac{2}{|\lambda|}.$$

Thus even when the differential equation is asymptotically stable, the step size must place every scaled eigenvalue $\tau\lambda$ inside the method's absolute stability region.

10. SEPTEMBER 16, 2026

We study the stability domains of explicit Runge–Kutta methods and use them to choose time steps. The examples show two different obstructions: rapidly decaying modes can force very small steps, and an undamped oscillator can become unstable under discretization. Throughout, A denotes the matrix in the differential equation; $\mathbf{A} = (a_{ij})$ denotes the Runge–Kutta coefficient matrix.

10.1. Continuous stability and discrete stability. For a constant matrix A , the solution of $x' = Ax$ is $x(t) = e^{tA}x(0)$. Recall the two spectral criteria:

	asymptotic stability	stability (boundedness)
$x' = Ax$	$\operatorname{Re} \lambda < 0$	$\operatorname{Re} \lambda \leq 0$
$x_{n+1} = Gx_n$	$ \mu < 1$	$ \mu \leq 1$

Here the conditions apply to every $\lambda \in \sigma(A)$ and $\mu \in \sigma(G)$, respectively. In the boundedness column, eigenvalues on the boundary must also be *semisimple*: every associated Jordan block must have size one. Algebraic multiplicity one is sufficient, but is not necessary.

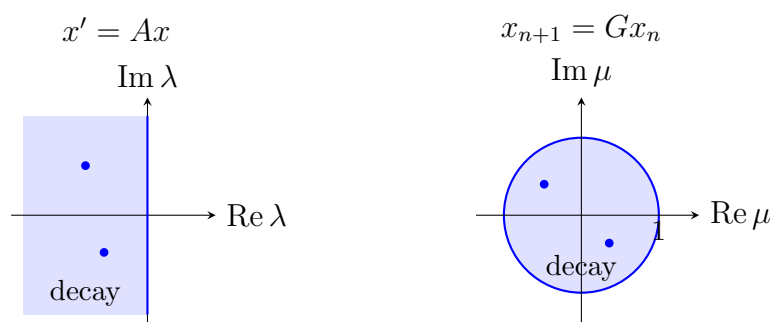


FIGURE 1. The open left half-plane governs continuous decay; the open unit disk governs discrete decay. Boundary eigenvalues require the additional Jordan-block condition for boundedness.

For an explicit RK method applied with a fixed step $\tau > 0$,

$$x_{n+1} = R(\tau A)x_n, \quad \sigma(R(\tau A)) = \{R(\tau\lambda) : \lambda \in \sigma(A)\}.$$

For example, the eigenvector identity $R(\tau A)v = R(\tau\lambda)v$ follows from $Av = \lambda v$. Triangularizing A proves the full spectral mapping identity even when A is not diagonalizable. Consequently,

$$\rho(R(\tau A)) = \max_{\lambda \in \sigma(A)} |R(\tau\lambda)|,$$

and strict inequality $\rho(R(\tau A)) < 1$ characterizes asymptotic stability of the numerical evolution.

10.2. The stability domain in the complex plane.

Definition 8 (Stability domain). *For the scalar test equation $y' = \lambda y$, put $z = \tau\lambda$ and write $y_{n+1} = R(z)y_n$. We use the notation*

$$(83) \quad \mathcal{D} = \{z \in \mathbb{C} : |R(z)| \leq 1\}, \quad \mathcal{S} = \{z \in \mathbb{C} : |R(z)| < 1\}.$$

Thus $\mathcal{D}$ describes bounded scalar iterates, and $\mathcal{S}$ describes decay. For a rational stability function, these sets contain only points where the function is defined.

The handwritten notes use the non-strict convention $\mathcal{D}$; $\mathcal{S}$ agrees with the strict convention used in the previous lecture. In particular, $R(0) = 1$, so $0 \in \mathcal{D}$ but $0 \notin \mathcal{S}$. To reproduce decay of a linear system,

every scaled eigenvalue $\tau\lambda$ must belong to $\mathcal{S}$. Membership in $\mathcal{D}$ alone does not settle a matrix problem with defective boundary eigenvalues.

10.2.1. *Forward Euler.* Here $R_1(z) = 1 + z$. Writing $z = x + iy$,

$$|R_1(z)| \leq 1 \iff (x+1)^2 + y^2 \leq 1.$$

The domain is the closed disk of radius one centered at -1 . For a negative real eigenvalue,

$$(84) \quad |1 + \tau\lambda| < 1 \iff 0 < \tau < \frac{2}{|\lambda|}.$$

At the upper endpoint the amplification factor is -1 : the scalar solution alternates without decaying.

10.2.2. *Two, three, and four stages.* An explicit two-stage method of order two has

$$R_2(z) = 1 + z + \frac{z^2}{2}.$$

Its boundary can be plotted using

$$\left(1 + x + \frac{x^2 - y^2}{2}\right)^2 + y^2(1 + x)^2 = 1.$$

On the negative real axis, $R_2(-r) = 1 - r + r^2/2$ is positive and is less than one exactly when $0 < r < 2$. Thus its real stability interval is the same as forward Euler's, despite its higher order.

The standard three-stage, third-order method and classical RK4 have

$$R_3(z) = 1 + z + \frac{z^2}{2} + \frac{z^3}{6}, \quad R_4(z) = 1 + z + \frac{z^2}{2} + \frac{z^3}{6} + \frac{z^4}{24}.$$

Their strict negative-real stability intervals are approximately

$$(-2.512745, 0) \quad \text{and} \quad (-2.785294, 0),$$

respectively. The following plots are computed from $|R_j(z)| = 1$, rather than schematic ovals.

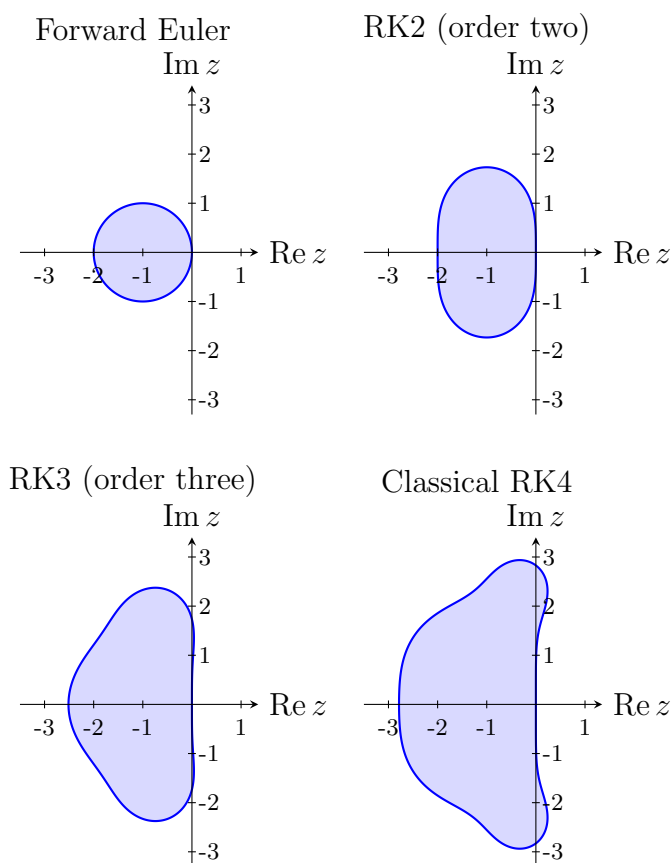


FIGURE 2. Computed stability domains for four standard explicit RK methods, on equal scales. Shading indicates $|R(z)| \leq 1$. Higher order changes the shape, but every one of these domains is bounded.

Remark 2 (Stages are not the same as order). An explicit s -stage method has a stability polynomial of degree *at most* s . If its order is p , then

$$R(z) = \sum_{j=0}^p \frac{z^j}{j!} + \sum_{j=p+1}^s \alpha_j z^j.$$

The degree can be smaller than s , and the coefficients beyond order p depend on the method. The formula $R(z) = \sum_{j=0}^s z^j/j!$ applies when the method has order s ; it does not describe every s -stage RK method.

10.3. Why every explicit RK stability domain is bounded. The statement “the stability domain is finite” means that it is *bounded*, not

that it contains finitely many points. The following completes the polynomial argument indicated in class.

Theorem 6. *Fix any finite number of stages $s \geq 1$. Every consistent explicit s -stage Runge–Kutta method has a bounded stability domain $\mathcal{D}$. In particular, no such method is A -stable.*

Proof. Let $\mathbf{1} = (1, \dots, 1)^\top$. On $y' = \lambda y$, the stage vector satisfies

$$(I - z\mathbf{A})k = \lambda y_n \mathbf{1}, \quad R(z) = 1 + zb^\top(I - z\mathbf{A})^{-1}\mathbf{1}.$$

Since $\mathbf{A}$ is strictly lower triangular, $\mathbf{A}^s = 0$. The finite geometric-series identity gives

$$(I - z\mathbf{A})^{-1} = I + z\mathbf{A} + \dots + z^{s-1}\mathbf{A}^{s-1},$$

for every complex z . Therefore

$$(85) \quad R(z) = 1 + \sum_{j=1}^s (b^\top \mathbf{A}^{j-1} \mathbf{1}) z^j.$$

Consistency gives $b^\top \mathbf{1} = 1$, so $R'(0) = 1$. In particular, R is not constant. Let its actual degree be m , where $1 \leq m \leq s$, and write

$$R(z) = \alpha_m z^m + \dots + \alpha_1 z + \alpha_0, \quad \alpha_m \neq 0.$$

Suppose, for a contradiction, that $\mathcal{D}$ is unbounded. There are $z_n \in \mathcal{D}$ with $|z_n| \rightarrow \infty$. On one hand,

$$\left| \frac{R(z_n)}{z_n^m} \right| \leq \frac{1}{|z_n|^m} \rightarrow 0.$$

On the other hand, division of the polynomial by z_n^m gives

$$\frac{R(z_n)}{z_n^m} = \alpha_m + \sum_{j=0}^{m-1} \frac{\alpha_j}{z_n^{m-j}} \rightarrow \alpha_m \neq 0,$$

a contradiction. Hence $\mathcal{D}$ is bounded.

For an explicit radius bound, set

$$C = \sum_{j=0}^{m-1} |\alpha_j|, \quad B = \max \left\{ 1, \frac{C+1}{|\alpha_m|} \right\}.$$

If $r = |z| > B$, the reverse triangle inequality yields

$$|R(z)| \geq |\alpha_m|r^m - Cr^{m-1} = r^{m-1}(|\alpha_m|r - C) > 1.$$

Thus $\mathcal{D} \subseteq \{|z| \leq B\}$. It is also closed by continuity of R , hence compact. Finally, an A -stable method must contain the entire open left half-plane in its stability domain. This is impossible for a bounded set. $\square$

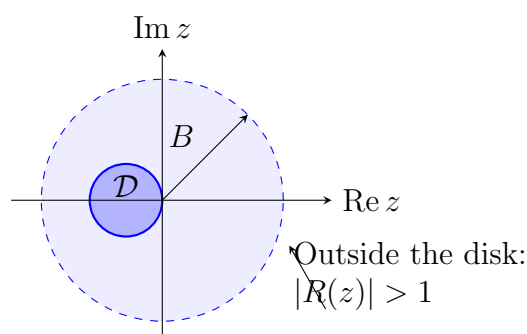


FIGURE 3. The polynomial's leading term dominates in every complex direction. The small region is schematic; the enclosing radius depends on the method's coefficients and need not be sharp.

Remark 3 (Scope of the theorem). The radius may change with s and with the RK coefficients; the theorem does not give one common bound for all explicit methods. Consistency rules out the degenerate constant polynomial $R \equiv 1$, whose non-strict domain would be all of $\mathbb{C}$. Implicit methods generally have rational stability functions and can have unbounded domains, as the next lecture shows.

10.4. A system with a fast decaying mode. Consider the example from class,

$$(86) \quad x' = Ax, \quad A = \begin{bmatrix} 0 & 1 \\ -50 & -51 \end{bmatrix}.$$

Its characteristic polynomial is

$$\det(\lambda I - A) = \lambda^2 + 51\lambda + 50 = (\lambda + 1)(\lambda + 50).$$

With eigenvectors $v_1 = (1, -1)^\top$ and $v_2 = (1, -50)^\top$, the exact solution is

$$x(t) = c_1 e^{-t} v_1 + c_2 e^{-50t} v_2.$$

Both modes decay, but on very different time scales: 1 and 1/50. Forward Euler must satisfy both

$$|1 - \tau| < 1, \quad |1 - 50\tau| < 1,$$

so its strict stability restriction is

$$(87) \quad \boxed{0 < \tau < \frac{1}{25} = 0.04.}$$

At $\tau = 0.04$, the fast numerical mode has multiplier -1 and fails to decay. Since this matrix is diagonalizable, the endpoint still gives bounded linear iterates. For $\tau > 0.04$, the fast mode is unstable. The second-order, two-stage explicit method has exactly the same restriction because its real stability interval is also $(-2, 0)$.

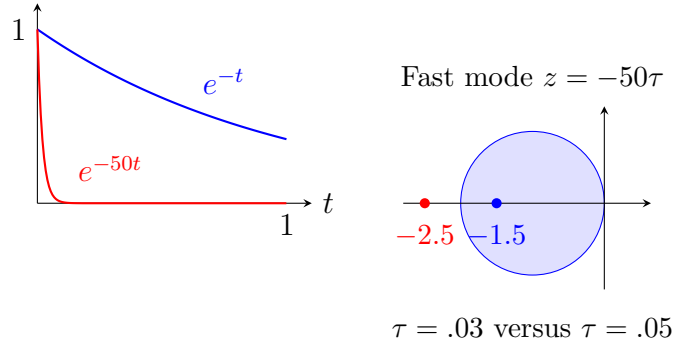


FIGURE 4. The fast exact mode quickly disappears, but it still limits Euler's step size. For $\tau = .03$, its scaled eigenvalue is inside the disk; for $\tau = .05$, it is outside.

This is the mechanism of *stiffness*: stability requires a much smaller step than is needed to resolve the slowly varying solution of interest. Even after the fast exact mode becomes negligible, roundoff and other perturbations can excite its unstable numerical counterpart. Stiffness depends on time scales and the integration interval, not merely on the numerical size of a dimensional eigenvalue.

10.5. The same restriction near a nonlinear equilibrium. The nonlinear example in the notes is

$$(88) \quad x' = f(x), \quad f(x_1, x_2) = \begin{bmatrix} x_2 \\ -51x_2 - 25(x_1^3 - x_1) \end{bmatrix}.$$

At $x_* = (1, 0)$, $f(x_*) = 0$, and

$$Df(x) = \begin{bmatrix} 0 & 1 \\ -25(3x_1^2 - 1) & -51 \end{bmatrix}, \quad Df(1, 0) = \begin{bmatrix} 0 & 1 \\ -50 & -51 \end{bmatrix}.$$

The equilibrium is locally asymptotically stable because the Jacobian has eigenvalues $-1, -50$.

Euler's map is $\Psi^\tau(x) = x + \tau f(x)$. It fixes x_* , and

$$D\Psi^\tau(x_*) = I + \tau Df(x_*).$$

For $0 < \tau < 1/25$, both multipliers are strictly inside the unit circle, so the discrete equilibrium is locally asymptotically stable. The same conclusion holds for explicit RK2, since the derivative of its map at an equilibrium is $R_2(\tau Df(x_*))$. At the boundary, linearization alone does not determine nonlinear stability.

As an additional illustration, write $q = x_1$, $v = x_2$. The system is a damped particle in the potential

$$U(q) = \frac{25}{4}(q^2 - 1)^2, \quad E(q, v) = \frac{1}{2}v^2 + U(q), \quad \frac{dE}{dt} = -51v^2 \leq 0.$$

The two potential wells correspond to $(\pm 1, 0)$; the equilibrium $(0, 0)$ is a saddle because its Jacobian has negative determinant.

10.6. An oscillator: forward Euler is never stable. For

$$(89) \quad x' = Ax, \quad A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix},$$

the eigenvalues are $\pm i$. Since $A^\top = -A$,

$$\frac{d}{dt} \|x(t)\|_2^2 = 0.$$

The exact trajectories are circles, and the origin is stable but not asymptotically stable. Forward Euler has eigenvalues $1 \pm i\tau$, whose

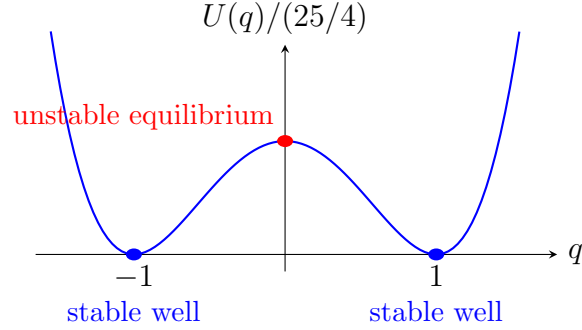


FIGURE 5. The potential of the nonlinear example. Damping decreases mechanical energy; the Jacobian gives the local decay rates near a well.

modulus is $\sqrt{1 + \tau^2} > 1$ for every $\tau > 0$. Equivalently,

$$(I + \tau A)^T(I + \tau A) = (1 + \tau^2)I, \quad \|x_n\|_2 = (1 + \tau^2)^{n/2} \|x_0\|_2.$$

Thus every positive fixed step causes artificial growth over long integration times. This does not contradict convergence as $\tau \rightarrow 0$ on a fixed finite time interval.

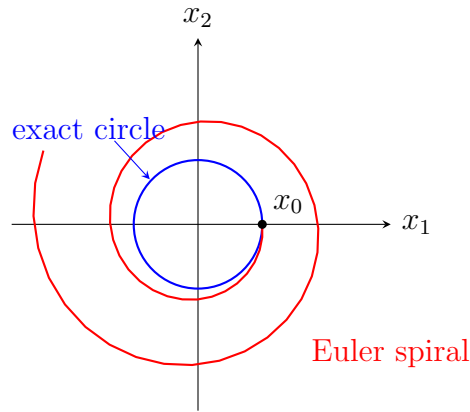


FIGURE 6. Forward Euler on the oscillator, starting at $(1, 0)$, with $\tau = .2$ for 50 steps. The exact orbit has radius one; the numerical radius grows by $\sqrt{1 + \tau^2}$ at each step.

For comparison, direct calculation gives

$$\begin{aligned} |R_2(iy)|^2 &= 1 + \frac{y^4}{4}, \\ |R_3(iy)|^2 &= 1 + \frac{y^4(y^2 - 3)}{36}, \\ |R_4(iy)|^2 &= 1 + \frac{y^6(y^2 - 8)}{576}. \end{aligned}$$

Hence RK2 is also unstable on this oscillator for every $\tau > 0$. RK3 and RK4 have bounded iterates for $0 < \tau \leq \sqrt{3}$ and $0 < \tau \leq 2\sqrt{2}$, respectively. Inside these intervals their amplification factors have modulus less than one, so they introduce numerical damping. Stability alone does not imply preservation of the exact oscillation amplitude or phase.

11. SEPTEMBER 18, 2026

We continue with stability domains, then introduce implicit methods whose domains contain the entire left half-plane. We retain the notation x_n , $\tau_n = t_{n+1} - t_n$, and the stage slopes k_i from September 2. For stability calculations the step is fixed, $\tau_n = \tau > 0$. As on September 16, A is the ODE matrix and $\mathbf{A} = (a_{ij})$ is the RK coefficient matrix (denoted by A in the earlier lectures devoted solely to RK coefficients).

11.1. Step-size restrictions for complex eigenvalues. For the linear system $x' = Ax$, a Runge–Kutta method gives

$$x_{n+1} = Gx_n, \quad G = R(\tau A), \quad \rho(G) = \max_{\lambda \in \sigma(A)} |R(\tau \lambda)|.$$

Strict inequality $\rho(G) < 1$ implies $G^n \rightarrow 0$. If $\rho(G) = 1$, boundedness additionally requires every unit-modulus eigenvalue of G to be semisimple.

For forward Euler and $\lambda = a + ib$ with $a < 0$,

$$\begin{aligned} |1 + \tau \lambda|^2 &= (1 + \tau a)^2 + \tau^2 b^2 \\ &= 1 + 2\tau a + \tau^2 |\lambda|^2. \end{aligned}$$

Thus

$$(90) \quad |1 + \tau\lambda| < 1 \iff 0 < \tau < \frac{-2 \operatorname{Re} \lambda}{|\lambda|^2}.$$

Increasing τ moves $z = \tau\lambda$ outward along a ray; the admissible step ends where that ray leaves the stability disk.

Example 16 (A damped oscillation). *The matrix in class is*

$$A = \begin{bmatrix} 1 & 2 \\ -4 & -3 \end{bmatrix}, \quad \det(\lambda I - A) = \lambda^2 + 2\lambda + 5.$$

Its eigenvalues are $\lambda_{1,2} = -1 \pm 2i$. Euler's condition is

$$|1 + \tau(-1 \pm 2i)|^2 = (1 - \tau)^2 + 4\tau^2 = 1 - 2\tau + 5\tau^2 < 1.$$

Therefore the numerical solution decays exactly when

$$0 < \tau < \frac{2}{5}.$$

At $\tau = 2/5$, the multipliers are $3/5 \pm (4/5)i$. They are distinct and have modulus one, so the iterates are bounded but do not decay. For larger steps the origin is unstable.

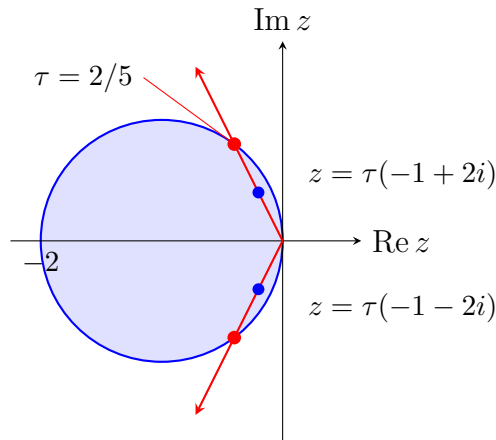


FIGURE 7. The scaled eigenvalues travel along conjugate rays. The points for $\tau = 1/5$ lie inside Euler's disk; the red points for $\tau = 2/5$ lie on its boundary.

For stiff equations, these restrictions can require many steps even when the solution of interest changes slowly. This motivates methods with no step-size restriction for decay of the linear test equation.

11.2. Backward Euler. Backward Euler evaluates the slope at the new time and state:

$$(91) \quad x_{n+1} = x_n + \tau f(t_{n+1}, x_{n+1}).$$

For $f(t, x) = Ax$,

$$(I - \tau A)x_{n+1} = x_n, \quad G_{\text{BE}} = (I - \tau A)^{-1}.$$

Why is the inverse defined for every $\tau > 0$ if A is Hurwitz? If $I - \tau A$ were singular, there would be a nonzero v with $Av = \tau^{-1}v$. This would give a positive real eigenvalue of A , a contradiction.

The rational spectral mapping identity gives

$$(92) \quad \sigma(G_{\text{BE}}) = \left\{ \frac{1}{1 - \tau\lambda} : \lambda \in \sigma(A) \right\}.$$

For completeness, if $B = I - \tau A$ is invertible, its eigenvalues are $1 - \tau\lambda$, and those of B^{-1} are their reciprocals. This also follows by triangularizing A ; no diagonalizability assumption is needed.

Thus the stability function and its non-strict domain are

$$(93) \quad R_{\text{BE}}(z) = \frac{1}{1 - z}, \quad \mathcal{D}_{\text{BE}} = \{z : |z - 1| \geq 1\}.$$

The domain is the exterior, including the boundary, of the unit disk centered at 1. The pole $z = 1$ is excluded automatically. For $\text{Re } z < 0$,

$$|1 - z|^2 = 1 - 2\text{Re } z + |z|^2 > 1,$$

so $|R_{\text{BE}}(z)| < 1$. Therefore, for every Hurwitz matrix A and every fixed $\tau > 0$, backward Euler is asymptotically stable. This is a statement about stability; resolving a solution accurately can still require a small step.

11.3. The implicit trapezoidal method. The trapezoidal rule averages the old and new slopes:

$$(94) \quad x_{n+1} = x_n + \frac{\tau}{2} (f(t_n, x_n) + f(t_{n+1}, x_{n+1})).$$

This is the *implicit* trapezoidal method. Heun's method, the explicit trapezoidal method from September 2, instead evaluates the second slope at a forward Euler prediction.

For $x' = Ax$, rearranging gives

$$\left(I - \frac{\tau}{2}A\right) x_{n+1} = \left(I + \frac{\tau}{2}A\right) x_n, \quad G_{\text{TR}} = \left(I - \frac{\tau}{2}A\right)^{-1} \left(I + \frac{\tau}{2}A\right).$$

If A is Hurwitz, the inverse exists for every $\tau > 0$ by the same argument as for backward Euler. Its eigenvalues are

$$\sigma(G_{\text{TR}}) = \left\{ \frac{1 + \tau\lambda/2}{1 - \tau\lambda/2} : \lambda \in \sigma(A) \right\}.$$

Hence

$$(95) \quad R_{\text{TR}}(z) = \frac{1 + z/2}{1 - z/2} = \frac{2 + z}{2 - z}.$$

Writing $z = x + iy$,

$$\begin{aligned} |R_{\text{TR}}(z)| \leq 1 &\iff |2 + z|^2 \leq |2 - z|^2 \\ &\iff (2 + x)^2 + y^2 \leq (2 - x)^2 + y^2 \\ &\iff x \leq 0. \end{aligned}$$

Thus

$$(96) \quad \mathcal{D}_{\text{TR}} = \{z : \operatorname{Re} z \leq 0\}, \quad \mathcal{S}_{\text{TR}} = \{z : \operatorname{Re} z < 0\}.$$

The pole $z = 2$ lies outside these sets. The trapezoidal method is also asymptotically stable for every positive fixed step when A is Hurwitz.

Definition 9 (*A-stability*). *A method is A-stable if its stability function is defined and satisfies $|R(z)| \leq 1$ throughout the open left half-plane.*

Backward Euler and the trapezoidal method are *A-stable*. Theorem 6 shows that no consistent explicit RK method with finitely many stages can be *A-stable*.

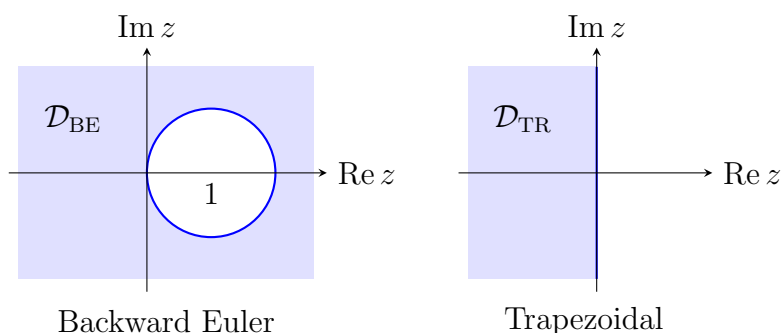


FIGURE 8. The shaded stability domains extend to infinity. Both contain the entire left half-plane; backward Euler also damps some modes in the right half-plane, which is not faithful to the growth of those exact modes.

11.4. Oscillations: growth, damping, or constant amplitude.

Return to the skew-symmetric oscillator matrix $A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ from (89). The exact norm is constant. The scalar amplification factors on the imaginary axis satisfy

method	$ R(i\tau) $	amplitude behavior
forward Euler	$\sqrt{1 + \tau^2}$	growth
backward Euler	$(1 + \tau^2)^{-1/2}$	decay
trapezoidal	1	constant

For backward Euler,

$$\|x_n\|_2 = (1 + \tau^2)^{-n/2} \|x_0\|_2;$$

the damping is artificial for this conservative problem.

For the trapezoidal method, put $q = \tau/2$. Since $A^2 = -I$,

$$G_{\text{TR}} = \frac{(1 - q^2)I + 2qA}{1 + q^2}, \quad G_{\text{TR}}^\top G_{\text{TR}} = I.$$

Thus the numerical update is an orthogonal rotation and preserves $\|x_n\|_2$ exactly. More generally, the same orthogonality follows for every real skew-symmetric matrix A , since the factors $I \pm qA$ commute.

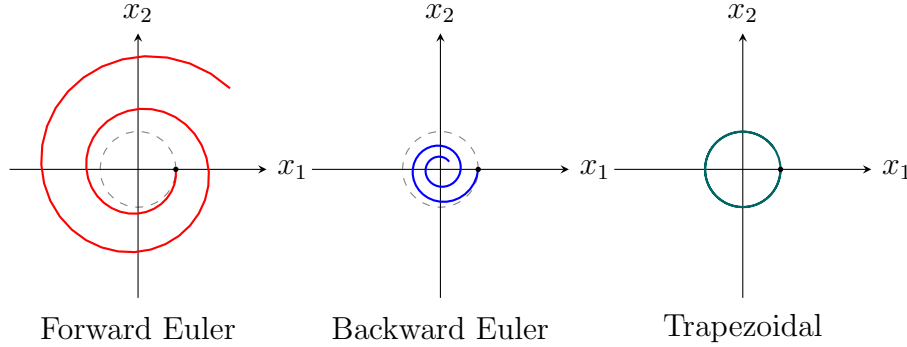


FIGURE 9. The same oscillator and initial value $(1, 0)$, with $\tau = .2$ for 60 steps in each panel, on identical scales. The dashed circle is the exact orbit. Trapezoidal vertices remain on the circle; the straight segments only join successive iterates.

Amplitude preservation does not imply exact phase. On the scalar mode $y' = iy$,

$$R_{\text{TR}}(i\tau) = \exp\left(2i \arctan \frac{\tau}{2}\right), \quad 2 \arctan \frac{\tau}{2} = \tau - \frac{\tau^3}{12} + O(\tau^5).$$

The exact phase advance is τ . The numerical frequency is

$$\frac{2}{\tau} \arctan \frac{\tau}{2}.$$

For the real oscillator drawn here, the rotation is clockwise, with this same angle magnitude. Purely imaginary eigenvalues alone do not imply conservation of the Euclidean norm for an arbitrary matrix; skew-symmetry is the property used in this example.

11.5. Damping very stiff modes: an additional comparison.

Both implicit methods allow arbitrary steps for linear stability, but they behave differently when $z = -r$ with $r \gg 1$:

$$R_{\text{BE}}(-r) = \frac{1}{1+r} \rightarrow 0, \quad R_{\text{TR}}(-r) = \frac{1-r/2}{1+r/2} \rightarrow -1.$$

The exact factor e^{-r} is very small. Backward Euler strongly damps such a mode, while the trapezoidal method can produce slowly decaying sign alternations when the step is large compared with the mode's decay time.

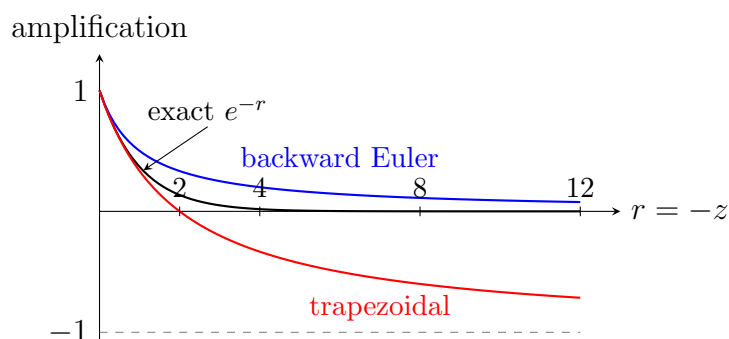


FIGURE 10. Signed amplification factors along the negative real axis. The trapezoidal factor changes sign at $r = 2$ and approaches -1 ; backward Euler's factor approaches zero.

An A -stable method is called L -stable if in addition $R(z) \rightarrow 0$ as $|z| \rightarrow \infty$ in the left half-plane. Thus backward Euler is L -stable and the trapezoidal method is not. This additional notion explains their different behavior on very fast decaying modes.

11.6. Implicit Runge–Kutta methods and their tableaux. The stage and update formulas are the same as in (26)–(27):

$$k_i = f \left(t_n + c_i \tau, x_n + \tau \sum_{j=1}^s a_{ij} k_j \right), \quad i = 1, \dots, s,$$

$$x_{n+1} = x_n + \tau \sum_{i=1}^s b_i k_i.$$

For an implicit method, the stage equations need not be solvable by successive explicit evaluations: a stage may depend on itself or on other unknown stages. They generally require solving algebraic systems, nonlinear if f is nonlinear. The internal consistency relation remains $c = A\mathbf{1}$.

11.6.1. Backward Euler as a one-stage RK method. Take $a_{11} = b_1 = c_1 = 1$:

$$\begin{array}{c|c} 1 & 1 \\ \hline & 1 \end{array}.$$

Then

$$k_1 = f(t_n + \tau, x_n + \tau k_1), \quad x_{n+1} = x_n + \tau k_1,$$

which is exactly (91). The coefficient $a_{11} = 1$ makes the sole stage implicit.

11.6.2. *Trapezoidal rule as a two-stage RK method.* Its tableau is

$$\begin{array}{c|cc} 0 & 0 & 0 \\ 1 & \frac{1}{2} & \frac{1}{2} \\ \hline & \frac{1}{2} & \frac{1}{2} \end{array}, \quad \mathbf{A} = \begin{bmatrix} 0 & 0 \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}.$$

Indeed,

$$\begin{aligned} k_1 &= f(t_n, x_n), \\ k_2 &= f\left(t_n + \tau, x_n + \frac{\tau}{2}k_1 + \frac{\tau}{2}k_2\right), \\ x_{n+1} &= x_n + \frac{\tau}{2}(k_1 + k_2). \end{aligned}$$

The argument of f in the second stage is x_{n+1} , so this recovers (94). Its weights satisfy $b^\top \mathbf{1} = 1$ and $b^\top c = 1/2$, giving order two by the earlier order conditions. In contrast, backward Euler has $b^\top c = 1$, so it has only order one.

11.7. **Derivation of the general RK stability function.** On $y' = \lambda y$, each stage satisfies

$$k_i = \lambda y_n + z \sum_{j=1}^s a_{ij} k_j, \quad z = \tau \lambda.$$

Collecting the slopes into $k = (k_1, \dots, k_s)^\top$ gives

$$(I - z\mathbf{A})k = \lambda y_n \mathbf{1}.$$

Whenever $I - z\mathbf{A}$ is invertible, substitution into the update produces

$$\begin{aligned} y_{n+1} &= y_n + \tau b^\top k \\ &= (1 + z b^\top (I - z\mathbf{A})^{-1} \mathbf{1}) y_n. \end{aligned}$$

Thus the formula from September 4 applies to both explicit and implicit RK schemes:

$$(97) \quad \boxed{R(z) = 1 + zb^T(I - zA)^{-1}\mathbf{1}.}$$

For implicit methods the inverse is generally a rational matrix function, so R need not be a polynomial. The determinant lemma also gives

$$R(z) = \frac{\det(I - zA + z\mathbf{1}b^T)}{\det(I - zA)}.$$

The numerator and denominator have degree at most s , before possible cancellations. At a singular stage matrix, the original stage equations must still be checked; a removable singularity in a simplified scalar formula does not by itself make the stages uniquely solvable.

For backward Euler this yields $R(z) = 1 + z/(1 - z) = 1/(1 - z)$. For the trapezoidal tableau, solve

$$(I - zA)w = \mathbf{1} \implies w_1 = 1, \quad w_2 = \frac{1 + z/2}{1 - z/2}.$$

Then $R(z) = 1 + z(w_1 + w_2)/2 = (1 + z/2)/(1 - z/2)$, as above. Implicitness permits A -stability but does not guarantee it; the actual rational stability function must be examined.

11.8. Connection with collocation. The lecture lists collocation as the next topic. The following short construction explains its connection with the implicit RK formulas. Choose distinct nodes $c_1, \dots, c_s \in [0, 1]$ and let u be a polynomial of degree at most s on one time step such that

$$u(t_n) = x_n, \quad u'(t_n + c_i\tau) = f(t_n + c_i\tau, u(t_n + c_i\tau)).$$

Let $\ell_j(\theta)$ be the Lagrange basis at these nodes and set $k_j = u'(t_n + c_j\tau)$. Interpolation of u' , followed by integration, gives

$$u(t_n + \theta\tau) = x_n + \tau \sum_{j=1}^s \left(\int_0^\theta \ell_j(\xi) d\xi \right) k_j.$$

Evaluating at the nodes and at $\theta = 1$ produces the RK coefficients

$$(98) \quad a_{ij} = \int_0^{c_i} \ell_j(\xi) d\xi, \quad b_j = \int_0^1 \ell_j(\xi) d\xi.$$

The single node $c_1 = 1$ gives backward Euler. The two nodes $0, 1$ give $\ell_1(\theta) = 1 - \theta$ and $\ell_2(\theta) = \theta$, hence exactly the trapezoidal tableau above.

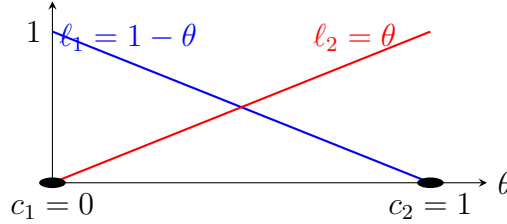


FIGURE 11. The endpoint Lagrange basis for trapezoidal collocation. Each basis function has integral $1/2$, giving the two equal weights; its integral up to each node gives the corresponding tableau row.

12. SEPTEMBER 21, 2026

We continue the study of implicit Runge–Kutta methods, derive their stability functions, and develop the collocation construction outlined on September 18. The main connection is that integrating an interpolating polynomial produces both the RK coefficients and an associated quadrature rule.

We retain x_n , $\tau_n = t_{n+1} - t_n$, and the stage slopes k_i . On a single step, write $\tau = \tau_n$. As in the previous lecture, $A = (a_{ij})$ denotes the RK coefficient matrix.

12.1. Implicit Runge–Kutta methods and stability functions.

An s -stage RK method has the form

$$(99) \quad k_i = f \left(t_n + c_i \tau, x_n + \tau \sum_{j=1}^s a_{ij} k_j \right), \quad i = 1, \dots, s,$$

$$(100) \quad x_{n+1} = x_n + \tau \sum_{i=1}^s b_i k_i.$$

The coefficients are arranged in the Butcher tableau

$$\begin{array}{c|c} c & \mathbf{A} \\ \hline & b^\top \end{array}, \quad c = \begin{bmatrix} c_1 \\ \vdots \\ c_s \end{bmatrix}, \quad b = \begin{bmatrix} b_1 \\ \vdots \\ b_s \end{bmatrix}.$$

For an implicit method, the stage equations generally form a coupled algebraic system. They are nonlinear when f is nonlinear.

Apply the method to the scalar test equation

$$x' = \lambda x, \quad \lambda \in \mathbb{C}, \quad z = \tau \lambda.$$

The stage equations become

$$k_i = \lambda x_n + z \sum_{j=1}^s a_{ij} k_j.$$

With $k = (k_1, \dots, k_s)^\top$ and $\mathbf{1} = (1, \dots, 1)^\top$, this is

$$(I - z\mathbf{A})k = \lambda x_n \mathbf{1}.$$

Whenever $I - z\mathbf{A}$ is invertible,

$$k = \lambda x_n (I - z\mathbf{A})^{-1} \mathbf{1}.$$

Substitution into the update gives

$$(101) \quad x_{n+1} = R(z)x_n, \quad \boxed{R(z) = 1 + zb^\top(I - z\mathbf{A})^{-1}\mathbf{1}.}$$

Thus an implicit RK method generally has a rational stability function.

For an explicit method, $\mathbf{A}^s = 0$, so

$$(I - z\mathbf{A})^{-1} = I + z\mathbf{A} + \dots + z^{s-1}\mathbf{A}^{s-1},$$

and R is a polynomial of degree at most s .

Keeping the distinction from the previous lectures, the strict absolute stability region and the non-strict stability domain are

$$\mathcal{S} = \{z \in \mathbb{C} : I - z\mathbf{A} \text{ is invertible, } |R(z)| < 1\},$$

$$\mathcal{D} = \{z \in \mathbb{C} : I - z\mathbf{A} \text{ is invertible, } |R(z)| \leq 1\}.$$

For a fixed step, $x_n = R(z)^n x_0$. Hence points in $\mathcal{S}$ give decay to zero, while points with $|R(z)| = 1$ give bounded scalar iterates without

decay. The invertibility condition ensures that the stages themselves are uniquely defined.

12.2. The collocation polynomial. Consider a system $x' = f(t, x)$, where the solution takes values in $\mathbb{R}^d$. Choose distinct nodes $c_1, \dots, c_s \in [0, 1]$. Let $\mathcal{P}_m$ denote the scalar polynomials of degree at most m .

Definition 10 (Collocation on one time step). *Seek a vector polynomial $u \in (\mathcal{P}_s)^d$ on $[t_n, t_{n+1}]$ satisfying*

$$(102) \quad u(t_n) = x_n,$$

$$(103) \quad u'(t_n + c_i\tau) = f(t_n + c_i\tau, u(t_n + c_i\tau)), \quad i = 1, \dots, s.$$

The next numerical value is

$$x_{n+1} = u(t_{n+1}).$$

The polynomial agrees with the initial value and satisfies the differential equation at the chosen collocation points. It need not satisfy the differential equation between those points. The initial condition and the s collocation conditions give $(s+1)d$ scalar equations for the $(s+1)d$ polynomial coefficients. For smooth f , the corresponding stage equations have a unique local solution near $\tau = 0$; arbitrary large steps need not have this property.

12.3. Lagrange interpolation and the RK coefficients. The Lagrange basis for the nodes is

$$(104) \quad \ell_j(\theta) = \prod_{\substack{m=1 \\ m \neq j}}^s \frac{\theta - c_m}{c_j - c_m}, \quad \ell_j(c_i) = \delta_{ij}.$$

Each ℓ_j has degree at most $s-1$. Given scalar data $\varphi(c_j)$, the polynomial

$$P(\theta) = \sum_{j=1}^s \varphi(c_j) \ell_j(\theta)$$

interpolates φ at all the nodes. In particular, if $\varphi \in \mathcal{P}_{s-1}$, then $P = \varphi$ identically.

Define the collocation slopes by

$$k_j = u'(t_n + c_j\tau) = f(t_n + c_j\tau, u(t_n + c_j\tau)).$$

Since u' has degree at most $s - 1$, its interpolation is exact:

$$(105) \quad u'(t_n + \theta\tau) = \sum_{j=1}^s k_j \ell_j(\theta).$$

Integrating and using the initial condition gives

$$(106) \quad \begin{aligned} u(t_n + \theta\tau) - x_n &= \tau \int_0^\theta u'(t_n + \xi\tau) d\xi \\ &= \tau \sum_{j=1}^s \left(\int_0^\theta \ell_j(\xi) d\xi \right) k_j. \end{aligned}$$

The factor τ comes from the change of variables $t = t_n + \xi\tau$.

At $\theta = c_i$, equation (106) yields

$$u(t_n + c_i\tau) = x_n + \tau \sum_{j=1}^s a_{ij} k_j, \quad a_{ij} = \int_0^{c_i} \ell_j(\theta) d\theta.$$

Inserting these stage values into (103) gives exactly the RK stage equations (99). At $\theta = 1$, we obtain

$$x_{n+1} = x_n + \tau \sum_{j=1}^s b_j k_j, \quad b_j = \int_0^1 \ell_j(\theta) d\theta.$$

Consequently, the collocation coefficients are

$$(107) \quad \boxed{a_{ij} = \int_0^{c_i} \ell_j(\theta) d\theta, \quad b_j = \int_0^1 \ell_j(\theta) d\theta.}$$

Conversely, slopes satisfying this RK system define a collocation polynomial through (106). Thus the two formulations are equivalent.

Because $\sum_j \ell_j(\theta) = 1$, the coefficients automatically satisfy

$$\sum_{j=1}^s a_{ij} = c_i, \quad \sum_{j=1}^s b_j = 1.$$

These are the internal consistency relation $c = \mathbf{A}\mathbf{1}$ and the first-order condition $b^\top \mathbf{1} = 1$.

Collocation generally produces implicit RK methods: the integrated basis functions couple the unknown stages. The one-node choice $c_1 = 0$ is an exception, giving forward Euler. The choices $c_1 = 1$ and $(c_1, c_2) = (0, 1)$ recover backward Euler and the implicit trapezoidal method, respectively, as on September 18.

12.4. The associated quadrature rule. Integrating the Lagrange interpolant of a function φ gives the interpolatory quadrature rule

$$(108) \quad \int_0^1 \varphi(\theta) d\theta \approx \sum_{j=1}^s b_j \varphi(c_j), \quad b_j = \int_0^1 \ell_j(\theta) d\theta.$$

The c_j are the quadrature nodes and the b_j are its weights. Write the quadrature error as

$$E[\varphi] = \int_0^1 \varphi(\theta) d\theta - \sum_{j=1}^s b_j \varphi(c_j).$$

We use E here to distinguish the error functional from the stability function $R(z)$.

Definition 11 (Degree of precision). *A quadrature rule has degree of precision p if*

$$E[\varphi] = 0 \quad \text{for every } \varphi \in \mathcal{P}_p,$$

but $E[\varphi] \neq 0$ for at least one $\varphi \in \mathcal{P}_{p+1}$.

Every interpolatory rule with these s distinct nodes has $p \geq s - 1$. Indeed, for $\varphi \in \mathcal{P}_{s-1}$,

$$\varphi(\theta) = \sum_{j=1}^s \varphi(c_j) \ell_j(\theta),$$

and integrating proves exactness. Equivalently, exactness through degree p is expressed by the moment conditions

$$(109) \quad \sum_{j=1}^s b_j c_j^m = \frac{1}{m+1}, \quad m = 0, \dots, p.$$

For $m = 0$, the left-hand side means $\sum_j b_j$.

The largest possible degree of precision for s real nodes is $2s - 1$. To see the upper bound, consider

$$q(\theta) = \prod_{j=1}^s (\theta - c_j)^2.$$

This polynomial has degree $2s$, vanishes at every node, and has strictly positive integral on $[0, 1]$. Therefore no such rule can be exact for every polynomial of degree $2s$. Gaussian quadrature attains the bound $p = 2s - 1$ with a suitable choice of nodes. Its use in RK methods is the topic announced for the next lecture.

12.5. Quadrature precision and collocation order.

Theorem 7 (Order of a collocation method). *Suppose the interpolatory quadrature rule associated with the collocation nodes has degree of precision p . Then the corresponding collocation RK method has order $p + 1$, assuming sufficient smoothness of f and the solution.*

Here p denotes quadrature precision; the RK order is $p + 1$. Thus a step started from the exact solution satisfies

$$x(t_{n+1}) - u(t_{n+1}) = O(\tau^{p+2}).$$

Under the usual smoothness and Lipschitz assumptions, this leads to global error $O(h^{p+1})$ on a fixed finite interval, where $h = \max_n \tau_n$. The order statement concerns the values at step endpoints; it does not assert the same error order at every point of the collocation polynomial.

The role of quadrature is especially transparent when $f(t, x) = g(t)$ is independent of x . Then

$$x_{n+1} = x_n + \tau \sum_{j=1}^s b_j g(t_n + c_j \tau),$$

whereas the exact increment is

$$x(t_{n+1}) - x(t_n) = \tau \int_0^1 g(t_n + \theta \tau) d\theta.$$

Taylor expansion of g and the moment conditions (109) give a one-step error of $O(\tau^{p+2})$. The theorem extends this order conclusion to general nonlinear collocation equations.

Example 17 (One collocation node). *With $s = 1$ and $c_1 = c$, the basis polynomial is $\ell_1 = 1$, so the tableau is*

$$\begin{array}{c|c} c & c \\ \hline & 1 \end{array}.$$

The quadrature rule is $\int_0^1 \varphi \approx \varphi(c)$. It is exact for linear polynomials precisely when $c = 1/2$. Thus the midpoint choice gives the second-order implicit midpoint method

$$x_{n+1} = x_n + \tau f \left(t_n + \frac{\tau}{2}, \frac{x_n + x_{n+1}}{2} \right).$$

The endpoint choices $c = 0$ and $c = 1$ have precision zero and give the first-order forward and backward Euler methods.

Example 18 (Endpoint collocation). *For $c_1 = 0$, $c_2 = 1$, the weights are $b_1 = b_2 = 1/2$. The quadrature rule is exact for constants and linear polynomials, but*

$$\frac{1}{2}(0^2 + 1^2) = \frac{1}{2} \neq \frac{1}{3} = \int_0^1 \theta^2 d\theta.$$

Hence $p = 1$, and the associated trapezoidal collocation method has order two.

For Gaussian nodes, $p = 2s - 1$, so the resulting s -stage Gauss collocation method has order $2s$. In particular, one Gauss node gives order two and two Gauss nodes give order four. The placement of the nodes, not just their number, determines the quadrature precision and hence the collocation order.

13. SEPTEMBER 23, 2026

We explain why quadrature precision determines the endpoint order of a collocation method, then construct Gauss Runge–Kutta methods. Exactness of Gaussian quadrature also gives preservation of quadratic

first integrals and a direct proof of A -stability. We retain $\tau = \tau_n$, the stage slopes k_i , and the RK matrix $\mathbf{A}$; A is reserved for an ODE matrix.

13.1. Collocation and quadrature error. Recall that the collocation polynomial $u \in (\mathcal{P}_s)^d$ satisfies

$$\begin{aligned} u(t_n) &= x_n, \\ u'(t_n + c_i\tau) &= f(t_n + c_i\tau, u(t_n + c_i\tau)), \quad i = 1, \dots, s, \\ x_{n+1} &= u(t_n + \tau). \end{aligned}$$

The associated RK coefficients are

$$a_{ij} = \int_0^{c_i} \ell_j(\theta) d\theta, \quad b_j = \int_0^1 \ell_j(\theta) d\theta,$$

where ℓ_j is the Lagrange basis from (104).

On the physical time interval, define the quadrature error functional

$$(110) \quad E_{n,\tau}[\varphi] = \int_{t_n}^{t_n+\tau} \varphi(t) dt - \tau \sum_{j=1}^s b_j \varphi(t_n + c_j\tau).$$

This is a linear functional. As on September 21, E denotes quadrature error, while $R(z)$ denotes a stability function.

Lemma 2 (Scaling of quadrature error). *Suppose the quadrature rule has degree of precision p . For $\varphi \in C^{p+1}([t_n, t_n + \tau])$,*

$$|E_{n,\tau}[\varphi]| \leq C\tau^{p+2} \max_{t_n \leq t \leq t_n+\tau} |\varphi^{(p+1)}(t)|,$$

where C depends only on the fixed nodes and weights.

Proof. Expand $\varphi(t_n + \tau\theta)$ through degree p :

$$\varphi(t_n + \tau\theta) = \sum_{m=0}^p \frac{\varphi^{(m)}(t_n)}{m!} \tau^m \theta^m + r(\theta),$$

with

$$|r(\theta)| \leq \frac{\tau^{p+1}}{(p+1)!} \max |\varphi^{(p+1)}|, \quad 0 \leq \theta \leq 1.$$

Exactness cancels every displayed polynomial term. The remaining integral and weighted sum are bounded by $\tau(1 + \sum_j |b_j|) \max |r|$, proving the estimate. $\square$

The same estimate holds componentwise for vector-valued functions. If the integrand itself depends on τ , the derivative bound must be uniform as $\tau \rightarrow 0$.

13.2. The collocation defect and variation of constants. For the local error analysis, let $x(t)$ be the exact solution started from $x(t_n) = x_n$. Define the defect

$$(111) \quad h(t) = u'(t) - f(t, u(t)).$$

Then the polynomial solves a perturbed differential equation,

$$u' = f(t, u) + h(t), \quad u(t_n) = x_n,$$

and the collocation conditions say exactly that

$$(112) \quad h(t_n + c_i \tau) = 0, \quad i = 1, \dots, s.$$

For the linear problem $x' = Ax$, variation of constants gives

$$u(t) = e^{(t-t_n)A} x_n + \int_{t_n}^t e^{(t-\eta)A} h(\eta) d\eta.$$

Thus, writing $T = t_n + \tau$,

$$(113) \quad x(T) - u(T) = - \int_{t_n}^T e^{(T-t)A} h(t) dt.$$

The minus sign follows from defining $h = u' - f(t, u)$ and taking the local error to be exact minus numerical, as in (19).

For a nonlinear equation, the analogous identity uses the exact flow $\Phi^{T,t}(y)$, namely the solution at time T started from y at time t . Put

$$M(T, t) = D_y \Phi^{T,t}(u(t)).$$

The flow identity $\partial_t \Phi^{T,t}(y) = -D_y \Phi^{T,t}(y) f(t, y)$ gives

$$\frac{d}{dt} \Phi^{T,t}(u(t)) = M(T, t) h(t).$$

Integrating from t_n to T yields the nonlinear variation-of-constants formula

$$(114) \quad x(T) - u(T) = - \int_{t_n}^T M(T, t) h(t) dt.$$

Set $\varphi(t) = M(T, t)h(t)$. By (112), every quadrature sample vanishes, so

$$x(T) - u(T) = -E_{n,\tau}[\varphi].$$

For sufficiently smooth f , on the local branch of collocation solutions for small τ , the required derivatives of φ remain bounded. The quadrature error estimate therefore gives

$$(115) \quad \boxed{x(t_n + \tau) - u(t_n + \tau) = O(\tau^{p+2}).}$$

This proves the order $p + 1$ conclusion stated on September 21. The extra accuracy at the endpoint comes from cancellation in the integral of the defect; merely bounding $\|h\|$ would miss that cancellation.

13.3. Gauss collocation. Gaussian quadrature with s nodes has degree of precision $2s - 1$, the largest possible value. Its nodes are the roots of the shifted Legendre polynomial of degree s :

$$L_s(\theta) = P_s(2\theta - 1), \quad 0 < c_1 < \cdots < c_s < 1,$$

where P_s is the Legendre polynomial on $[-1, 1]$. Equivalently, the monic nodal polynomial $\pi_s(\theta) = \prod_{j=1}^s (\theta - c_j)$ is orthogonal to $\mathcal{P}_{s-1}$ on $[0, 1]$:

$$\int_0^1 \pi_s(\theta)q(\theta) d\theta = 0, \quad q \in \mathcal{P}_{s-1}.$$

To see the resulting precision, divide any $v \in \mathcal{P}_{2s-1}$ as $v = \pi_s q + r$, with $q, r \in \mathcal{P}_{s-1}$. The integral of $\pi_s q$ vanishes by orthogonality, and all its quadrature samples vanish as well. The interpolatory rule integrates r exactly. Hence it integrates v exactly.

The nodes and weights satisfy

$$(116) \quad c_j = 1 - c_{s+1-j}, \quad b_j = b_{s+1-j} > 0.$$

Positivity follows directly from exactness applied to ℓ_j^2 , whose degree is $2s - 2$:

$$b_j = \sum_{i=1}^s b_i \ell_j(c_i)^2 = \int_0^1 \ell_j(\theta)^2 d\theta > 0.$$

Collocation at these nodes gives the *Gauss RK method* with s stages. The quadrature precision is $2s - 1$, so (115) gives

$$\text{RK order} = 2s, \quad \text{local error} = O(\tau^{2s+1}).$$

13.3.1. *One stage: implicit midpoint.* For $s = 1$, $c_1 = 1/2$, $b_1 = 1$, and $a_{11} = 1/2$:

$$\begin{array}{c|c} \frac{1}{2} & \frac{1}{2} \\ \hline & 1 \end{array}.$$

Thus

$$k_1 = f\left(t_n + \frac{\tau}{2}, x_n + \frac{\tau}{2}k_1\right), \quad x_{n+1} = x_n + \tau k_1.$$

Equivalently,

$$x_{n+1} = x_n + \tau f\left(t_n + \frac{\tau}{2}, \frac{x_n + x_{n+1}}{2}\right).$$

This method has order two. Its stability function is

$$R_{G,1}(z) = \frac{1 + z/2}{1 - z/2}.$$

It agrees with the trapezoidal stability function on the linear test equation, although the two methods differ for general nonlinear f .

13.3.2. *Two-stage Gauss method.* The roots of $L_2(\theta) = 6\theta^2 - 6\theta + 1$ are

$$c_1 = \frac{1}{2} - \frac{\sqrt{3}}{6}, \quad c_2 = \frac{1}{2} + \frac{\sqrt{3}}{6}.$$

Integrating their Lagrange basis functions gives

$$(117) \quad \begin{array}{c|cc} \frac{1}{2} - \frac{\sqrt{3}}{6} & \frac{1}{4} & \frac{1}{4} - \frac{\sqrt{3}}{6} \\ \frac{1}{2} + \frac{\sqrt{3}}{6} & \frac{1}{4} + \frac{\sqrt{3}}{6} & \frac{1}{4} \\ \hline & \frac{1}{2} & \frac{1}{2} \end{array}.$$

In particular, the upper-right entry has a minus sign, so that the first row sums to c_1 . The stage equations are

$$\begin{aligned} k_1 &= f \left(t_n + c_1 \tau, x_n + \tau \left[\frac{1}{4} k_1 + \left(\frac{1}{4} - \frac{\sqrt{3}}{6} \right) k_2 \right] \right), \\ k_2 &= f \left(t_n + c_2 \tau, x_n + \tau \left[\left(\frac{1}{4} + \frac{\sqrt{3}}{6} \right) k_1 + \frac{1}{4} k_2 \right] \right), \\ x_{n+1} &= x_n + \frac{\tau}{2} (k_1 + k_2). \end{aligned}$$

Both slopes must be solved for together. The method has order four, and substitution into (101) gives

$$(118) \quad R_{G,2}(z) = \frac{1 + z/2 + z^2/12}{1 - z/2 + z^2/12}.$$

13.4. Quadratic first integrals. For the autonomous equation $x' = f(x)$, a function $I(x)$ is a *first integral* if it is constant along every solution. Consider a quadratic function

$$I(x) = x^\top Q x, \quad Q^\top = Q.$$

Positive definiteness is not needed for conservation. Since

$$\frac{d}{dt} I(x(t)) = 2x(t)^\top Q f(x(t)),$$

the condition for a quadratic first integral is

$$(119) \quad x^\top Q f(x) = 0 \quad \text{for every } x.$$

Theorem 8 (Quadratic invariants of Gauss methods). *Every Gauss collocation method preserves each quadratic first integral exactly, provided its stage equations are solved exactly.*

Proof. Let $U(\theta) = u(t_n + \tau\theta)$ and $U_j = U(c_j)$. The polynomial U has degree at most s , and

$$U'(c_j) = \tau f(U_j).$$

Therefore $2U^\top Q U'$ has degree at most $2s - 1$ and Gaussian quadrature is exact for it. Consequently,

$$I(x_{n+1}) - I(x_n) = \int_0^1 2U(\theta)^\top Q U'(\theta) d\theta$$

$$= 2\tau \sum_{j=1}^s b_j U_j^\top Q f(U_j) = 0.$$

The last equality uses (119) at each stage value. $\square$

For example, if $f(x) = Ax$ and $A^\top Q + QA = 0$, the quadratic form is preserved by the exact solution and by every Gauss method. Taking $Q = I$ and $A^\top = -A$ recovers norm preservation for the oscillator from September 18. The argument relies on the invariant being quadratic; it does not give exact preservation of a general nonlinear first integral.

13.5. An energy identity for the scalar test equation. Apply Gauss collocation to $x' = \lambda x$, with $z = \tau\lambda$. Now U is complex-valued and $U'(c_j) = zU_j$. Since the derivative of $|U|^2$ is a real polynomial of degree at most $2s - 1$, exact quadrature gives

$$\begin{aligned} |x_{n+1}|^2 - |x_n|^2 &= 2 \int_0^1 \operatorname{Re}(\overline{U(\theta)} U'(\theta)) d\theta \\ (120) \qquad \qquad &= 2 \operatorname{Re} z \sum_{j=1}^s b_j |U_j|^2. \end{aligned}$$

For $\operatorname{Re} z < 0$, positivity of the weights implies $|x_{n+1}| \leq |x_n|$.

This also proves unique solvability in the open left half-plane. Indeed, for a homogeneous stage system take $x_n = 0$. The left side is then nonnegative and the right side nonpositive, so every $U_j = 0$. Since $U(0) = 0$ and the Gauss nodes are distinct and interior, U has $s + 1$ distinct zeros and must vanish identically. Thus the stage matrix has trivial kernel.

For $x_n \neq 0$, the sum in (120) is strictly positive: if all $U_j = 0$, then all $U'(c_j) = 0$, so $U' \equiv 0$ and $U \equiv 0$, a contradiction. Therefore

$$\operatorname{Re} z < 0 \implies |R(z)| < 1.$$

Every Gauss RK method is consequently A -stable. On the imaginary axis, the same identity gives $|R(z)| = 1$ whenever the stages are defined, explaining the absence of artificial damping for oscillatory modes. We continue this comparison with Radau methods in the next lecture.

14. SEPTEMBER 25, 2026

We compare Gauss and Radau collocation through their effects on quadratic quantities. Gauss quadrature gives an exact energy identity; right Radau quadrature contributes an additional nonpositive term. This leads to stability and contractivity results, and explains why Radau methods can damp very stiff modes more effectively.

As before, $\tau = \tau_n$, $\mathbf{A} = (a_{ij})$, and $z = \tau\lambda$. We use the normalized collocation polynomial

$$U(\theta) = u(t_n + \tau\theta), \quad U_j = U(c_j), \quad 0 \leq \theta \leq 1.$$

Its derivative is with respect to θ , so $U'(c_j) = \tau f(t_n + c_j\tau, U_j)$.

14.1. Gauss stability revisited. On the test equation $x' = \lambda x$, the numerical update is $x_{n+1} = R(z)x_n$, and

$$|x_{n+1}|^2 - |x_n|^2 = (|R(z)|^2 - 1)|x_n|^2.$$

Thus absolute stability is equivalent to nonincrease of the squared modulus. For Gauss collocation, the energy identity from (120) is

$$(121) \quad |x_{n+1}|^2 - |x_n|^2 = 2 \operatorname{Re} z \sum_{j=1}^s b_j |U_j|^2.$$

There is no quadrature remainder: the integrand $2 \operatorname{Re}(\bar{U}U')$ has degree at most $2s - 1$, exactly the Gaussian degree of precision.

For nonzero initial data and a uniquely solvable stage system, the sum on the right is positive. Hence

$$\begin{array}{l|l} \operatorname{Re} z < 0 & |R(z)| < 1 \\ \operatorname{Re} z = 0 & |R(z)| = 1 \\ \operatorname{Re} z > 0 & |R(z)| > 1 \end{array}.$$

As proved on September 23, the stages are uniquely solvable throughout the open left half-plane. In the notation used throughout these notes, the strict region $\mathcal{S}$ is the open left half-plane, and the non-strict domain $\mathcal{D}$ also includes admissible points on the imaginary axis.

14.2. Right Radau quadrature. Right Radau quadrature fixes the last node at the endpoint,

$$0 < c_1 < \cdots < c_{s-1} < c_s = 1.$$

For $s \geq 2$, its interior nodes are the roots of the shifted Jacobi polynomial $P_{s-1}^{(1,0)}(2\theta - 1)$. Equivalently, the monic polynomial

$$\pi_{s-1}(\theta) = \prod_{j=1}^{s-1} (\theta - c_j)$$

is orthogonal to $\mathcal{P}_{s-2}$ with weight $1 - \theta$:

$$\int_0^1 (1 - \theta) \pi_{s-1}(\theta) q(\theta) d\theta = 0, \quad q \in \mathcal{P}_{s-2}.$$

For $s = 1$, take $\pi_0 = 1$ and the sole node $c_1 = 1$. The interpolatory weights are again $b_j = \int_0^1 \ell_j(\theta) d\theta$.

The degree of precision is $2s - 2$. Indeed, divide a polynomial of degree at most $2s - 2$ by $(\theta - 1)\pi_{s-1}(\theta)$. The remainder has degree at most $s - 1$, so it is integrated exactly by interpolation. The other term has zero integral by the weighted orthogonality and vanishes at every node. As for Gauss quadrature,

$$b_j = \int_0^1 \ell_j(\theta)^2 d\theta > 0,$$

because $\deg \ell_j^2 \leq 2s - 2$.

Collocation at these nodes gives the *Radau IIA* family. The endpoint choice $c_s = 1$ specifies the right Radau family used in this lecture. Its order is

$$(2s - 2) + 1 = 2s - 1.$$

14.3. The sign of the Radau quadrature error. On $[0, 1]$, let

$$E[\varphi] = \int_0^1 \varphi(\theta) d\theta - \sum_{j=1}^s b_j \varphi(c_j).$$

Define the positive constant

$$(122) \quad \kappa_s = \int_0^1 (1 - \theta) \pi_{s-1}(\theta)^2 d\theta = \frac{s((s-1)!)^4}{2((2s-1)!)^2} > 0.$$

The polynomial $(\theta - 1)\pi_{s-1}(\theta)^2$ is monic of degree $2s - 1$ and vanishes at every quadrature node. Its lower degree terms are integrated exactly, so

$$(123) \quad E[\theta^{2s-1}] = -\kappa_s.$$

In particular, precision cannot exceed $2s - 2$.

For a real function $\varphi \in C^{2s-1}([0, 1])$, the usual remainder formula is

$$(124) \quad E[\varphi] = -\frac{\kappa_s}{(2s-1)!}\varphi^{(2s-1)}(\xi), \quad \xi \in [0, 1].$$

One way to obtain it is to use the Hermite interpolant matching φ and φ' at each interior node and matching φ at 1. Its remainder contains the factor $(\theta - 1)\pi_{s-1}(\theta)^2$, which has one sign on the interval. Integration then gives (124). For the energy argument below, the polynomial identity (123) is already sufficient.

14.4. The Radau energy identity and A-stability. Write the complex scalar collocation polynomial as

$$U(\theta) = \alpha_s \theta^s + \cdots + \alpha_0.$$

The highest-degree term in the derivative of $|U|^2$ is

$$\frac{d}{d\theta}|U(\theta)|^2 = 2s|\alpha_s|^2\theta^{2s-1} + \text{terms of degree at most } 2s - 2.$$

Consequently its Radau quadrature error is $-2s\kappa_s|\alpha_s|^2$. On $x' = \lambda x$, $U'(c_j) = zU_j$, so

$$(125) \quad \boxed{|x_{n+1}|^2 - |x_n|^2 = 2 \operatorname{Re} z \sum_{j=1}^s b_j |U_j|^2 - 2s\kappa_s |\alpha_s|^2.}$$

Both terms on the right are nonpositive when $\operatorname{Re} z \leq 0$. Thus $|R(z)| \leq 1$, provided the stage system is solvable.

The identity also proves solvability in the closed left half-plane. For a homogeneous stage system, $U(0) = 0$. Equation (125) forces $U(1) = 0$ and $\alpha_s = 0$. If $\operatorname{Re} z < 0$, it also forces all stage values to vanish, so $U \equiv 0$. If $\operatorname{Re} z = 0$, the polynomial $U' - zU$ has degree at most $s - 1$ and vanishes at all s nodes. It is therefore identically zero. Together

with $U(0) = 0$, this again gives $U \equiv 0$. Hence the stage matrix has trivial kernel. Every Radau IIA method is A -stable.

14.5. Quadratic energy. For a real autonomous system, let

$$I(x) = x^\top Qx, \quad Q^\top = Q.$$

The exact derivative is

$$\frac{d}{dt}I(x(t)) = 2x(t)^\top Qf(x(t)).$$

Thus $x^\top Qf(x) = 0$ gives a first integral, while $x^\top Qf(x) \leq 0$ gives a nonincreasing quadratic quantity. Write $U(\theta) = \alpha_s \theta^s + \cdots + \alpha_0$, now with vector coefficients. For Gauss and Radau IIA, respectively, the quadrature arguments give

$$(126) \quad I(x_{n+1}) - I(x_n) = 2\tau \sum_{j=1}^s b_j U_j^\top Qf(U_j),$$

$$(127) \quad I(x_{n+1}) - I(x_n) = 2\tau \sum_{j=1}^s b_j U_j^\top Qf(U_j) - 2s\kappa_s \alpha_s^\top Q\alpha_s.$$

For Gauss methods, a quadratic first integral is preserved for any symmetric Q . For Radau methods, the last term is nonpositive when $Q \succeq 0$; it generally causes numerical dissipation even when the exact quantity is conserved. More generally, if

$$Q \succeq 0, \quad x^\top Qf(x) \leq 0 \quad \text{for all } x,$$

then both families satisfy

$$x_{n+1}^\top Qx_{n+1} \leq x_n^\top Qx_n.$$

The assumption $Q \succeq 0$ matters for the Radau conclusion: an indefinite quadratic form does not give a fixed sign for the remainder term.

14.6. One-stage and two-stage Radau IIA methods. For $s = 1$, the tableau is

$$\begin{array}{c|c} 1 & 1 \\ \hline & 1 \end{array},$$

so the method is backward Euler, of order one.

For $s = 2$, the nodes and weights are

$$c_1 = \frac{1}{3}, \quad c_2 = 1, \quad b_1 = \frac{3}{4}, \quad b_2 = \frac{1}{4}.$$

The Lagrange basis is

$$\ell_1(\theta) = \frac{3}{2}(1 - \theta), \quad \ell_2(\theta) = \frac{3}{2}\theta - \frac{1}{2}.$$

Integrating up to each node gives the tableau

$$(128) \quad \begin{array}{c|cc} \frac{1}{3} & \frac{5}{12} & -\frac{1}{12} \\ 1 & \frac{3}{4} & \frac{1}{4} \\ \hline & \frac{3}{4} & \frac{1}{4} \end{array}.$$

Hence

$$\begin{aligned} k_1 &= f\left(t_n + \frac{\tau}{3}, x_n + \tau \left[\frac{5}{12}k_1 - \frac{1}{12}k_2 \right]\right), \\ k_2 &= f\left(t_n + \tau, x_n + \tau \left[\frac{3}{4}k_1 + \frac{1}{4}k_2 \right]\right), \\ x_{n+1} &= x_n + \tau \left(\frac{3}{4}k_1 + \frac{1}{4}k_2 \right). \end{aligned}$$

The quadrature precision is two, so the method has order three. For example, the moments through degree two are

$$\frac{3}{4} + \frac{1}{4} = 1, \quad \frac{3}{4} \frac{1}{3} + \frac{1}{4} = \frac{1}{2}, \quad \frac{3}{4} \frac{1}{9} + \frac{1}{4} = \frac{1}{3}.$$

Its stability function is

$$(129) \quad R_{R,2}(z) = \frac{1 + z/3}{1 - 2z/3 + z^2/6}.$$

14.7. Dissipative ODEs and contractivity.

Definition 12 (Dissipative vector field). *The autonomous ODE $x' = f(x)$ is dissipative in the Euclidean inner product if*

$$(130) \quad \langle f(x) - f(y), x - y \rangle \leq 0 \quad \text{for all } x, y.$$

For two exact solutions,

$$\frac{d}{dt} \|x(t) - y(t)\|_2^2 = 2 \langle f(x(t)) - f(y(t)), x(t) - y(t) \rangle \leq 0.$$

Therefore

$$(131) \quad \|x(t) - y(t)\|_2 \leq \|x(t_0) - y(t_0)\|_2, \quad t \geq t_0.$$

This concerns the distance between any two solutions. If $f(0) = 0$, taking $y(t) = 0$ also gives nonincrease of $\|x(t)\|_2$.

Gauss and Radau IIA preserve the same contractivity property. To see this, let U, V be the collocation polynomials for starting values x_n, y_n on the same step, and put $W = U - V$. At the nodes,

$$W'(c_j) = \tau(f(U_j) - f(V_j)).$$

The Gauss energy identity applied to W gives

$$\|x_{n+1} - y_{n+1}\|_2^2 - \|x_n - y_n\|_2^2 = 2\tau \sum_{j=1}^s b_j \langle U_j - V_j, f(U_j) - f(V_j) \rangle \leq 0.$$

For Radau IIA, an additional term $-2s\kappa_s \|\alpha_s(U) - \alpha_s(V)\|_2^2$ appears, which is also nonpositive. Thus both methods satisfy

$$(132) \quad \boxed{\|x_{n+1} - y_{n+1}\|_2 \leq \|x_n - y_n\|_2}.$$

The conclusion holds whenever the stage equations have been solved. It imposes no additional step-size restriction for contractivity. This preservation of contractivity for dissipative nonlinear ODEs is commonly called *B-stability*.

14.8. Stiff accuracy and L -stability. An A -stable method is L -stable if

$$R(z) \longrightarrow 0 \quad \text{as } |z| \rightarrow \infty \text{ in the left half-plane.}$$

This additional condition suppresses modes whose decay times are much shorter than the chosen step.

Suppose $\mathbf{A}$ is invertible. For large $|z|$,

$$\begin{aligned} (I - z\mathbf{A})^{-1} &= -z^{-1}(\mathbf{A} - z^{-1}I)^{-1} \\ &= -z^{-1}\mathbf{A}^{-1} - z^{-2}\mathbf{A}^{-2} + O(|z|^{-3}). \end{aligned}$$

Substitution into the RK stability function gives

$$(133) \quad R(z) = 1 - b^T A^{-1} \mathbf{1} + O(|z|^{-1}).$$

Thus, for an A -stable method with invertible A , the additional condition for L -stability is

$$(134) \quad b^T A^{-1} \mathbf{1} = 1.$$

For Radau IIA, $c_s = 1$, so the coefficient formula gives

$$a_{sj} = \int_0^{c_s} \ell_j(\theta) d\theta = \int_0^1 \ell_j(\theta) d\theta = b_j.$$

The last row of A therefore equals b^T . This property, together with $c_s = 1$, is called *stiff accuracy*: the last stage value is the new solution,

$$U_s = x_n + \tau \sum_j a_{sj} k_j = x_{n+1}.$$

If e_s is the last coordinate vector, then

$$b^T = e_s^T A \implies b^T A^{-1} \mathbf{1} = e_s^T \mathbf{1} = 1.$$

For completeness, A is invertible for collocation at distinct nonzero nodes. If $Av = 0$, the polynomial

$$V(\theta) = \int_0^\theta \sum_{j=1}^s v_j \ell_j(\xi) d\xi$$

has degree at most s and vanishes at $0, c_1, \dots, c_s$. Hence $V \equiv 0$, so $v_j = V'(c_j) = 0$ for every j . This applies to both Gauss and Radau IIA nodes.

Combining invertibility, stiff accuracy, and A -stability proves that every Radau IIA method is L -stable. The endpoint condition $c_s = 1$ yields the row identity for collocation methods; it would not by itself imply stiff accuracy for an arbitrary RK tableau.

Gauss methods are not L -stable. Their invertible coefficient matrices ensure a finite limit in (133), while their unit modulus on the imaginary axis forces that limit to have modulus one. For the first two methods,

the difference is explicit:

method	$R(z)$	$\lim_{r \rightarrow \infty} R(-r)$
Gauss, $s = 1$	$\frac{1 + z/2}{1 - z/2}$	-1
Gauss, $s = 2$	$\frac{1 + z/2 + z^2/12}{1 - z/2 + z^2/12}$	1
Radau IIA, $s = 1$	$\frac{1}{1 - z}$	0
Radau IIA, $s = 2$	$\frac{1 + z/3}{1 - 2z/3 + z^2/6}$	0

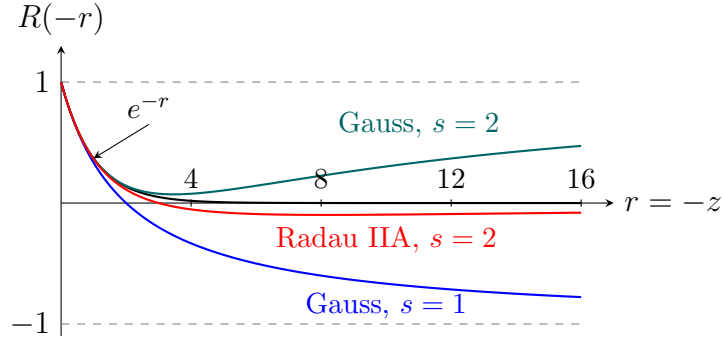


FIGURE 12. Amplification of a decaying scalar mode. All three numerical methods are A -stable, but the Radau factor tends to zero for very stiff modes. The Gauss factors eventually approach -1 and 1 , respectively.